

The Thiele Machine

A Complete Mathematical Specification

Derived from the Fully Machine-Checked Coq Corpus

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Before the specification. I don't do math by hand. I think in shapes, directions, and connections, and my part of this work was to see where things had to fit, to direct, and to refuse what didn't fit. This document is the translation. Language models wrote the formal statements and the proofs, Coq checked every one of them, and I read the statements to make sure they say what the shape says. So nothing below is in my voice, on purpose. It's the reference layer: exact statements, exact premises, and where each one lives. If you want the pictures, the monograph starts with them. If you want to know exactly what is claimed, it's here.

The Thiele Machine is an abstract model of computation, and an argument made with it about what an account of computation should preserve.

It is a machine in the sense a Turing machine is one: a mathematical object, not a device. The 51-opcode VM specified in the chapters below is one build of that model, specified in full so it can be checked. It is not the model. The model is the abstract **CertificationSystem** and **Substrate** interfaces; the VM is one instance of each.

Certification is one concrete witness of a broader investigation of structural state, observation, and cost. The abstract laws, theorems, and executable realizations belong to that investigation. Selected projections lose distinctions that the richer state can express; cost laws constrain specified events under explicit hypotheses.

A conventional encoding can preserve those distinctions. A projection that discards them cannot recover them in general. The stronger proposal that these distinctions belong in a necessary foundation of computation remains a research claim. Neither a physical cost metric nor infinitely many independent geometric dimensions follows from the existence of many observables. The status of each result is stated below.

Abstract

This is the complete mathematical specification of the Thiele Machine: a formal model of computation where structural information carries an explicit, monotone cost μ that starts at zero and never decreases. Everything here is built from the machine-checked Coq corpus. You can reconstruct the machine from what's in this document alone.

Coq establishes that conclusions follow from axioms. It does *not* establish that axioms model physical reality. That distinction is kept explicit throughout. Every result is tagged as one of three types: **(S)** structural theorems about the machine as a mathematical object, **(C)** conditional derivations where a physics conclusion follows from a stated axiom, and **(R)** consistency relations that check internal coherence but are not independent predictions.

The core claims are structural (S): No Free Insight; μ -conservation; μ -hierarchy; μ -ledger necessity (five-way independence classification); categorical separation; strict Turing subsumption of instruction sets; the classical CHSH bound ($|S| \leq 2$); the algebraic Tsirelson bound ($|S| \leq 2\sqrt{2}$); and the biconditional with no project-local axioms `column_contractive_iff_npa_psd`. The CHSH algebraic result is pure math. The Landauer connection is motivation. Geometry: the discrete angle-defect identity under its incidence assumptions and the additional restriction $B = 3\chi$.

The document also covers, with explicit epistemological tags, conditional derivations under named physical axioms (Born rule under linearity, no-cloning, unitarity) and reference explorations (4D simplicial-complex differential geometry, self-reference tower). These are marked (C) or (R) and do not form part of the primary claim ledger. Two named trust boundaries discussed here are: `bsc_kami_compilation_trusted` (Layer 2 Verilog semantics gap) and `A_QM` (physical quantum axiom).

Contents

1	Introduction	4
1.1	Epistemological Framework	5
2	Notation and Conventions	6
2.1	Overloaded symbols	6
2.2	Terms used before definition	7
3	The State Space	8
3.1	Primitives	8
3.2	Regions and Modules	8
3.3	The Partition Graph	8
3.4	Machine State	8
4	Instruction Set and Cost Model	9
4.1	The Axiom Scheme A1–A6	9
4.1.1	The consolidated <code>CertificationSystem</code> (A1–A2)	9
4.1.2	The <code>QuantitativeCertificationSystem</code> extension (A3–A6)	9
4.2	Instruction Categories	10
4.2.1	Structural Operations	10
4.2.2	Logical Operations	10
4.2.3	Computational Operations (Reversible ALU)	11
4.2.4	Standard ALU and Control Flow	11
4.2.5	Auxiliary Operations	11
4.2.6	Categorical Morphism Operations	11
4.3	Cost Model	12
4.4	The <code>CERTIFY</code> instruction	12
5	Operational Semantics	14
5.1	Transition Function	14
5.2	Traces and Execution	14
5.3	Selected Instruction Semantics	15
6	No Free Insight	16
6.1	Definitions	16
6.2	The Theorem	16
6.3	μ -Ledger Necessity	16
6.4	Partition Refinement and No Free Insight	18
7	The μ-Chaitin Theorem	19

8	Computational Universality and Subsumption	20
8.1	Strict Containment	20
8.2	Oracle Cost Lower Bound	20
8.3	μ -budget predicates	20
8.4	Intrinsic level hierarchy	21
8.5	Verifier model and the bare-setting impossibility	22
9	μ-Conservation and Receipt Integrity	24
9.1	The μ -Ledger Identity	24
9.2	Receipt Integrity	24
10	Gauge Shifts of the Ledger and a Noether-Style Invariance	25
10.1	The Partial $\mathbb{Z}$ -Action (Gauge Group)	25
10.2	Gauge Invariance	25
11	The Born Rule	27
12	Unitarity	28
13	Purification	29
14	No-Cloning	30
15	The Tsirelson Bound	31
16	Witness Insight Taxonomy	32
17	Bell's Theorem in Morphism Coupling Language	33
18	Slice Coherence and NPA Realizability: The Biconditional	34
18.1	The CHSH statistic S and its witness counts	34
18.2	The Elliptope Completion and Its Gate	35
18.3	The Pointer-Observable Criterion	36
18.3.1	The adversarial search	36
19	Spacetime Analogies: a μ-Pseudometric, Reachability Cones, and Discrete Geometry	39
19.1	The μ -Metric	39
19.2	Causal Cones	39
19.3	No-Signaling (Data Isolation)	39
19.4	Discrete Geometry: A Restricted Angle-Defect Identity	40
19.4.1	Cross-link in MuGravity Vocabulary	40
19.5	Four-Dimensional Discrete Riemannian Geometry (Reference Only, Not a Main Claim)	41
19.5.1	4D Simplicial Complex	41
19.5.2	Metric Tensor from μ -Costs	42
19.5.3	Discrete Differential Geometry	42
19.5.4	Curvature Tensors	43
19.5.5	Stress-Energy Tensor	43
19.5.6	Einstein's Field Equations	44
19.5.7	Empirical Validation	46
19.5.8	Affine Metric-Scaled Operator: Non-Uniform Boundary Results	46
20	Information Causality	48

21 Thermodynamic Bridge: Conditional Links and Their Limits	49
21.1 Landauer’s Principle	49
21.2 A2 from Landauer (Factored Form): a Conditional Whose Premises Fail on the Full ISA	49
21.3 Permanent Certification Forces a Merge	50
21.4 Exact Pricing Cannot Be Computed From a Window	52
21.5 Which Narrowing a Merge Price Charges	53
21.6 A Recursion Theorem for L	53
21.7 The Dimensional Gap: Why Ledgers Cannot Fix Constants	53
22 Self-Reference and the Thiele Manifold	55
22.1 Self-Reference Escalation	55
22.2 The Thiele Manifold	55
23 Three-Layer Observable Agreement	57
23.1 RTL Coverage: Zero Gaps	57
23.2 Named Trust Boundaries	58
24 Categorical Structure	59
24.1 The Morphism Layer	59
24.1.1 Objects and Morphisms	59
24.1.2 The Seven Categorical Opcodes	59
24.1.3 Category Laws	60
24.1.4 The Categorical Separation Theorem	60
25 Falsifiable Predictions	61
25.1 Falsify No Free Insight	61
25.2 Falsify the Classical CHSH Bound	61
25.3 Falsify the Biconditional	61
25.4 Falsify the μ -Hierarchy	61
25.5 Open Questions	61
A Proof File Index	63
B Epistemological Classification	66

Chapter 1

Introduction

The Thiele Machine makes the cost of structure explicit. Ordinary step-counting assigns one time unit per transition; it does not specify this structural-cost schedule. The Thiele Machine assigns a specific cost μ to every operation that extracts or asserts structural information. The chapters below lay out the consequences.

The central claim, proven formally: *specified certification and structural-entitlement events have the cost floors proved under their stated hypotheses*. Alongside that core, the specification records conditional results in which explicitly supplied premises (linearity, an information-conservation inequality, a second-law record field) yield statements with the shape of results in quantum mechanics and thermodynamics. In most of them μ enters as a named cost field, and the supplied premises do the work. They are marked **(C)**: conditional on those premises, not derived from the ledger alone. Each result’s status, whether a structural theorem **(S)**, a conditional derivation **(C)**, or a consistency relation **(R)**, is marked directly.

The document is organized as:

1. **Foundations:** State space, instruction set, operational semantics.
2. **Core Theorems:** No Free Insight, μ -ledger necessity, μ -Chaitin, μ -conservation.
3. **Computation:** Strict Turing subsumption of instruction sets, oracle pricing as instantiation of universal NFI, μ -complexity classes, μ -cost hierarchy.
4. **Symmetry & Conservation:** Gauge shifts of μ , a Noether-style invariance, receipts.
5. **Quantum / CHSH algebra:** Tsirelson bound (algebraic), CHSH classical bound, biconditional column-contractive $\iff$ NPA-realizable, three-tier observation taxonomy. Pure polynomial arithmetic, zero physics axioms.
6. **Conditional physics:** Born rule (under linearity), unitarity, no-cloning, purification, Landauer (with second-law assumed as a record field), affine EFE family. Each is conditional on stated axioms in its own file.
7. **Discrete Geometry:** Restricted angle-defect identity with $B = 3\chi$ (not classical Gauss-Bonnet); 4D simplicial complex and Einstein-tensor definitions (reference exploration only, vacuum case is a $0 = 0$ consistency check).
8. **Meta-Theory:** Three-layer Coq/Python/RTL observable contract, categorical structure.
9. **Predictions:** Falsifiable predictions tied to specific Coq-verifiable claims.

1.1 Epistemological Framework

Formal verification (Coq) establishes that conclusions follow logically from axioms. It does **not** establish that axioms model physical reality. This distinction is critical for evaluating the physics claims in Parts V–VIII.

Every result in this specification falls into one of three categories:

- (S) **Structural** Theorems about the Thiele Machine as a mathematical object. Their conclusions follow from the definitions, explicit hypotheses, and reported library assumptions. *Examples:* μ -conservation, gauge invariance, confluence, halting undecidability, strict Turing subsumption.
- (C) **Conditional** Theorems of the form “if axiom X holds, then physics result Y follows.” The logical inference is verified; whether axiom X holds in nature is an empirical question. *Examples:* Born rule uniqueness (conditional on linearity), complex necessity (conditional on 2D amplitudes), unitarity (conditional on information conservation).
- (R) **Consistency** Algebraic identities or definitional equivalences that verify internal coherence but do not constitute independent predictions. *Example:* the Tsirelson cost definition.

Each major theorem in Parts V–VIII is annotated with its category. A theorem marked (C) or (R) is not diminished. Conditional derivations and consistency checks are valuable scientific tools. But the reader should not mistake them for derivations from first principles without stated assumptions. See Appendix B for the complete classification table.

Chapter 2

Notation and Conventions

This chapter exists because several symbols in this document are overloaded, and at least one (S) is used well before it is defined. Read it once; it resolves ambiguities that would otherwise require guessing from context.

2.1 Overloaded symbols

S , **four distinct meanings**. Disambiguated by context and, where ambiguity would be real, by explicit qualification:

1. $\mathcal{S}$ (calligraphic): the *state space*.
2. S (roman, in operational-semantics chapters): a *particular VM state*, an element of $\mathcal{S}$.
3. $S(n)$ (applied to a natural number): the *successor function* on $\mathbb{N}$, i.e. $S(n) = n+1$. This is Coq's $\mathbf{S}$ constructor, and it is the reason cost expressions such as $\text{flen} \times 8 + S(\text{cost})$ are bounded below by 1: $S(n) \geq 1$ for every $n \in \mathbb{N}$. **This is the notation used from Chapter 4 onward, before its first formal appearance in the cost tables.**
4. S (in quantum chapters): the *CHSH statistic*, defined in Section 18.1. Real-valued.

G , **three distinct meanings**. All three occur, and two occur inside the same chapter on emergent geometry:

1. G : the *partition graph* component of a VM state.
2. G : *Newton's constant*, fixed to $1/(8\pi)$ as a unit choice.
3. $G_{\mu\nu}$: the *Einstein tensor*.

Where the geometry chapter needs the partition graph it is written G_{part} .

μ , **ledger and index, and two different number systems**.
1. μ : the *cost ledger*, a component of VM state. In the machine, μ **ranges over** $\mathbb{N}$: it is a natural number, it is monotone non-decreasing under the step relation, and A2 gives it an integer floor.
2. μ, ν as *tensor indices* in the geometry chapter ($g_{\mu\nu}, T_{\mu\nu}$). Unrelated to the ledger.
3. μ in the **quantum chapters** (measurement cost, the no-cloning inequality, the μ/I ratio) is **real-valued**. This is a genuinely different object from the machine's μ : there is no theorem in this development identifying them, and no coercion between them is claimed. Where the distinction matters the real-valued quantity is written $\mu_{\mathbb{R}}$.

2.2 Terms used before definition

CERTIFY. The instruction family whose step sets the certification bit. Formally $\text{CERTIFY}(\Delta\mu)$ has step semantics $\text{cert} \mapsto \text{true}$ with cost $S(\Delta\mu) \geq 1$; see Section 4.4. Chapters 5, 7 and 8 depend on it.

irreversible_bits. The count of bits whose prior value cannot be recovered from the post-state of a step. It is the quantity Landauer’s principle prices. Used in the thermodynamic corollaries.

has_structure_addition. At least one executed step within the supplied fuel changes the certification CSR from zero to non-zero, as defined in `coq/kernel/nfi/NoFreeInsight.v`.

Chapter 3

The State Space

3.1 Primitives

Let $\mathbb{N} = \{0, 1, 2, \dots\}$. Let $\mathbb{B} = \{0, 1\}$. Let Σ be a finite alphabet.

3.2 Regions and Modules

Definition 3.1 (Region). A **Region** R is a finite subset of $\mathbb{N}$, canonically represented as a deduplicated list preserving first-occurrence order: $\text{norm}(L) = \text{nodup}(L)$ removes duplicate elements while preserving the order of their first appearance. (The Python VM applies a stricter canonical form via `sorted(set(region))`; the Coq specification uses `nodup Nat.eq_dec` which does not sort.)

Definition 3.2 (Module). A **Module** $M = (R_M, A_M)$ consists of a normalized region $R_M \subset \mathbb{N}$ and a set of axioms $A_M \subset \Sigma^*$.

3.3 The Partition Graph

Definition 3.3 (Partition Graph). A partition graph $G = (N_{id}, \mathcal{T})$ where $N_{id} \in \mathbb{N}$ is the next available module ID and $\mathcal{T} : \{0, \dots, N_{id} - 1\} \rightarrow \mathcal{M}$ is a finite partial function from IDs to modules.

Axiom 3.1 (Well-Formedness). G is well-formed iff $\text{dom}(\mathcal{T}) \subseteq \{0, \dots, N_{id} - 1\}$.

3.4 Machine State

Definition 3.4 (VM State). The complete state is a 12-tuple:

$$S = (G, C, R, Mem, PC, \mu, T_\mu, err, \ell, m, w, cert)$$

where $G \in \mathcal{G}$ is the partition graph, $C \in \mathbb{N}^4$ comprises the CSRs (certificate address, status, error code, heap base), $R \in \mathbb{N}^{\text{REG_COUNT}}$ is the register file, $Mem \in \mathbb{N}^{\text{MEM_SIZE}}$ is main memory (the kernel constants `REG_COUNT` and `MEM_SIZE` are parametric; the synthesized RTL fixes `REG_COUNT = 16` and `MEM_SIZE = 128`), $PC \in \mathbb{N}$ is the program counter, $\mu \in \mathbb{N}$ is the μ -ledger, $T_\mu \in \mathbb{N}^{16}$ is the flattened 4×4 μ -tensor (row-major), $err \in \mathbb{B}$ is the global error flag (latching), $\ell \in \mathbb{N}$ is the logic-engine accumulator, $m \in \mathbb{N}$ is the machine status register (0 = Turing mode, 1 = Thiele mode), $w \in \mathbb{N}^8$ is the WitnessCount record (`vm_witness`): eight natural-number counters `wc_same_ab` and `wc_diff_ab` for each setting pair $(a, b) \in \{00, 01, 10, 11\}$, and $cert \in \mathbb{B}$ is the certified flag (`vm_certified`).

Chapter 4

Instruction Set and Cost Model

4.1 The Axiom Scheme A1–A6

“A2” is referred to throughout this document, and this is the scheme it belongs to. These are *conditions on Coq record types*, not **Axiom** declarations. The assumption receipt distinguishes project-local declarations from standard-library axioms and explicit theorem hypotheses. Supplying an instance of the record is the only way to invoke the corresponding theorems, and the 51-opcode VM supplies them by construction.

4.1.1 The consolidated **CertificationSystem** (A1–A2)

A **CertificationSystem** is a tuple $(\Sigma, \mathcal{I}, \text{step}, \text{cost}, \text{cert})$ where Σ is any state type, $\mathcal{I}$ any instruction type, $\text{step} : \Sigma \times \mathcal{I} \rightarrow \Sigma$, $\text{cost} : \mathcal{I} \rightarrow \mathbb{N}$, and $\text{cert} : \Sigma \rightarrow \mathbb{B}$, subject to:

A1 (certification is observable). A total function $\text{cert} : \Sigma \rightarrow \mathbb{B}$ exists. This holds *by record existence*: providing the field is providing the axiom.

A2 (certification is priced; the load-bearing one).

$$\forall s \in \Sigma, \forall i \in \mathcal{I}, \quad \text{cert}(s) = \text{false} \wedge \text{cert}(\text{step}(s, i)) = \text{true} \implies \text{cost}(i) \geq 1. \quad (4.1)$$

In Coq this is the field `cs_cert_costs`. Everything called “No Free Insight” in this document is A2 lifted from steps to traces by induction; the trace-level statement is genuinely proved, but its content is A2 plus “a sum of naturals containing a summand ≥ 1 is ≥ 1 ”.

Remark 4.1 (A2 is a stipulation, and that is the intended reading). A2 is a condition a substrate may or may not satisfy, not a discovered fact about computation. Nothing in this document derives it; Section 21.2 derives it *conditionally* from a Landauer premise plus a named calibration bridge, which relocates the stipulation rather than removing it. Claims of the form “the machine proves certification must cost” should be read as “on any substrate satisfying A2, certification costs”. That is what `universal_nfi_any_substrate` states, over an arbitrary state type and an arbitrary instruction type.

4.1.2 The **QuantitativeCertificationSystem** extension (A3–A6)

The quantitative extension lifts the floor from ≥ 1 to $\geq K$ for an arbitrary threshold. It adds a witness function $W : \Sigma \rightarrow \mathbb{N}$ and a threshold $K \in \mathbb{N}$, subject to:

A3 (cost bounds witness growth). $\forall s, i : W(s) + \text{cost}(i) \geq W(\text{step}(s, i))$. Evidence cannot accumulate faster than it is paid for. Coq: `qcs_cost_bounds_witness`.

A4 (certification requires the threshold). $\forall s : \text{cert}(s) = \text{true} \implies W(s) \geq K$. Coq: `qcs_cert_threshold_witness`.

A5 (clean start). The initial state is uncertified, $\text{cert}(s_0) = \text{false}$, and carries no witness, $W(s_0) = 0$. Supplied as a hypothesis at use sites rather than as a record field.

A6 (threshold is the claim’s price). K is the parameter of the conclusion: a run from s_0 to a certified state has total cost $\geq K$. Not an independent assumption; it names the quantity A3 and A4 telescope into.

Remark 4.2 (Two numbering systems, and which one this document uses). The kernel carries a second, older numbering attached to the `AbstractCertMachine` record, in which the cost-on-certification rule is called **A4** and **A2** names “cert starts unset”. The two systems agree on content and disagree on labels. **Throughout this document, “A2” means the cost-on-certification rule in the consolidated CertificationSystem sense stated above.** When reading kernel sources that mention `AbstractCertMachine`, translate accordingly.

4.2 Instruction Categories

The instruction set $\mathcal{I}$ is organized into six families. Every instruction i carries an explicit cost parameter $\Delta\mu_i \in \mathbb{N}$.

4.2.1 Structural Operations

These modify the partition graph G :

- **PNEW**($R, \Delta\mu$): Create module with region R .
- **PSPLIT**($m, L, R, \Delta\mu$): Split module m into disjoint regions.
- **PMERGE**($m_1, m_2, \Delta\mu$): Merge two modules.
- **PDISCOVER**($m, E, \Delta\mu$): Discovery accounting for module m . The abstract specification attaches evidence E (list of axioms) to the module; the current hardware-aligned step advances PC and charges $\Delta\mu$ without mutating the graph.
- **MDLACC**($m, \Delta\mu$): Minimum-description-length accumulate on module m .

4.2.2 Logical Operations

These interact with information content:

- **LASSERT**($freg, creg, kind, flen, cost$): Assert a formula read from `vm_mem[R[freg]]` (encoded-unit count $flen$, eight concrete bits per unit) against a witness package at `vm_mem[R[creg]]`. In SAT mode the package contains one satisfying assignment and one falsifying assignment, stored back-to-back; $\Delta\mu = flen \times 8 + S(cost) \geq 1$.
- **LJOIN**($c1reg, c2reg, \Delta\mu$): Join certificates from two memory-resident strings addressed by registers.
- **REVEAL**($m, n, cert, \Delta\mu$): Reveal n bits of structure from m ; cost is $n + S(\Delta\mu)$.
- **EMIT**($m, p, \Delta\mu$): Emit payload p from module m ; cost is $|p|_{\text{bits}} + S(\Delta\mu)$.

4.2.3 Computational Operations (Reversible ALU)

Reversible register/memory operations:

- $\text{XFER}(dst, src, \Delta\mu)$: Copy register src to dst .
- $\text{XOR_LOAD}(dst, addr, \Delta\mu)$: Load memory word at $addr$.
- $\text{XOR_ADD}(dst, src, \Delta\mu)$: $dst \leftarrow dst \oplus src$.
- $\text{XOR_SWAP}(a, b, \Delta\mu)$: Swap registers a and b .
- $\text{XOR_RANK}(dst, src, \Delta\mu)$: $dst \leftarrow \text{popcount}(src)$.

4.2.4 Standard ALU and Control Flow

- $\text{LOAD_IMM}(dst, imm, \Delta\mu)$: Load 32-bit immediate into register.
- $\text{LOAD}(dst, addr, \Delta\mu)$: Load from main memory.
- $\text{STORE}(addr, src, \Delta\mu)$: Store to main memory.
- $\text{ADD}(dst, r_1, r_2, \Delta\mu)$: $dst \leftarrow r_1 + r_2 \pmod{2^{64}}$.
- $\text{SUB}(dst, r_1, r_2, \Delta\mu)$: $dst \leftarrow r_1 - r_2 \pmod{2^{64}}$.
- $\text{JUMP}(target, \Delta\mu)$: Unconditional jump.
- $\text{JNEZ}(r, target, \Delta\mu)$: Jump if $R[r] \neq 0$.
- $\text{CALL}(target, \Delta\mu)$: Push return address, jump to target.
- $\text{RET}(\Delta\mu)$: Pop return address, jump back.

4.2.5 Auxiliary Operations

- $\text{CHSH_TRIAL}(x, y, a, b, \Delta\mu)$: Run a CHSH correlation trial; $x, y, a, b \in \{0, 1\}$ required.
- $\text{CHECKPOINT}(label, \Delta\mu)$: Execution checkpoint (charges μ).
- $\text{READ_PORT}(dst, ch, val, bits, \Delta\mu)$: Read from I/O channel (cert-setter).
- $\text{WRITE_PORT}(ch, src, \Delta\mu)$: Write to I/O channel.
- $\text{HEAP_LOAD}(dst, addr, \Delta\mu)$: Load from heap-relative address.
- $\text{HEAP_STORE}(addr, src, \Delta\mu)$: Store to heap-relative address.
- $\text{HALT}(\Delta\mu)$: Halt execution.

4.2.6 Categorical Morphism Operations

These 7 opcodes (0x27–0x2D) implement the categorical layer over partition modules. See Chapter 24 for laws and proofs.

- $\text{MORPH}(dst, src, tgt, idx, \Delta\mu)$: Create morphism $src \rightarrow tgt$; write morphism ID to $\text{regs}[dst]$.
- $\text{COMPOSE}(dst, f, g, \Delta\mu)$: Compose $f; g$ when $f.tgt = g.src$; errors on type mismatch.
- $\text{MORPH_ID}(dst, m, \Delta\mu)$: Create identity morphism id_m (diagonal relation).

- **MORPH_DELETE**($f, \Delta\mu$): Remove morphism f ; cascades when modules are split/merged.
- **MORPH_ASSERT**($f, prop, cert, \Delta\mu$): **Cert-setter**. Charges $S(\Delta\mu) \geq 1$ unconditionally. Even failed attempts (morphism does not exist) incur cost 1. This is the strongest form of the NoFI law: the machine refuses to let even a doomed attempt at certifying a relational claim slip through for free.
- **MORPH_TENSOR**($dst, f, g, \Delta\mu$): Tensor product $f \otimes g$ (requires union modules to exist).
- **MORPH_GET**($dst, f, sel, \Delta\mu$): Read morphism metadata: selector 0 = src, 1 = tgt, 2 = coupling length, 3 = is_identity.

A reminder before the machinery: this ISA is scaffolding. It is one way to build the substrate, not the substrate itself. Burn all 51 opcodes and the machine is exactly as real, because the machine is the law, not the instruction set.

The active implementation exposes a 51-opcode ISA: 14 arithmetic/data, 8 memory and control flow, 7 I/O and tensor, 4 partition operations, 3 logic and certification, 8 witness and certification (**CHSH_TRIAL**, **CERTIFY**, **REVEAL**, **CHSH_LASSERT**, and the four Q_{1+AB} check variants), 7 categorical morphism operations ($14 + 8 + 7 + 4 + 3 + 8 + 7 = 51$). 47 of the 51 are synthesized. The four Q_{1+AB} variants live in the Kami HW abstraction with kernel-equivalence proven; their hardware has not been built, so they are not in the synthesized Verilog. This condensed mathematical specification presents the core instruction families relevant to the kernel semantics and cost model. Every instruction carries an explicit $\Delta\mu$ parameter; the function `instruction_cost` extracts it.

4.3 Cost Model

Every instruction i carries an explicit cost parameter $\Delta\mu_i \in \mathbb{N}$. The function `instruction_cost(i)` extracts this parameter. The cost is applied via `apply_cost(s, i) = s. $\mu$  + instruction_cost(i)`. Since $\Delta\mu_i : \mathbb{N}$, the μ -ledger is monotonically non-decreasing by construction.

The mandatory positive-cost class is: **CERTIFY**, **MORPH_ASSERT**, and **LJOIN** (each charges $S(\Delta\mu) \geq 1$); **EMIT** ($|payload|_{\text{bits}} + S(\Delta\mu)$); **REVEAL** and **READ_PORT** ($bits + S(\Delta\mu)$); and **LASSERT** ($flen \times 8 + S(\Delta\mu)$). These cannot be zero-cost even with $\Delta\mu = 0$. The five **CHSH_LASSERT** family instructions also charge $S(\Delta\mu)$. Other instructions charge their encoded $\Delta\mu$, which may be zero.

Theorem 4.1 (Least joint unit floor, (**S**)). *For a finite family of event predicates on instructions, the least natural-valued schedule satisfying a unit floor on each event is the indicator that at least one event occurs. This is `joint_floor_is_least`. Independent Boolean coordinates can change together for one unit under these rules; additivity requires an additional condition.*

4.4 The CERTIFY instruction

CERTIFY is referred to throughout Chapters 6 onward and is defined here.

Definition 4.1 (**CERTIFY** (`instr_certify`, `VMStep.v`)). **CERTIFY**($\Delta\mu$) has the step semantics

$$pc \mapsto pc + 1, \quad \mu \mapsto \mu + S(\Delta\mu), \quad cert \mapsto \text{true}, \quad (4.2)$$

with every other state component unchanged. Its cost is $S(\Delta\mu) \geq 1$, which is the concrete instance of A2 (Section 4.1) for this ISA. It is the *only* rule in `VMStep.v` that sets `cert := true`.

Remark 4.3 (**CERTIFY** is ungarded; what that does and does not mean). The rule fires for every state and every $\Delta\mu$: it carries no precondition, takes no proposition as an argument, and checks nothing. Setting the certification bit is therefore always *available*, at a price of at least

1. Conversely, the instruction that does perform a decidable check on data, namely `LASSERT`, which traps on failure, leaves both the certification CSR and `cert` unchanged.

The honest description of the mechanism is therefore *metering*, not verification: the step rule guarantees that flipping the bit is priced, not that the bit is only set when some claim holds. Statements of the form “this machine makes stamping *verified* impossible” overstate it; what is impossible is stamping it *for free*.

The ISA does contain genuinely guarded instructions. `MORPH_ASSERT` requires the named morphism to exist, and `CHSH_LASSERT` runs `column_contractive_check_witness` on actual witness data and traps on failure, with a soundness bridge to PSD. A substrate in which the certification bit may be set only when an invariant holds would let A2 be *derived* (the cost of setting the bit becomes the cost of establishing the invariant) rather than stipulated. That is an open direction, not a present result; see Remark 4.1.

Chapter 5

Operational Semantics

5.1 Transition Function

The transition $\delta : \mathcal{S} \times \mathcal{I} \rightarrow \mathcal{S}$ is defined by:

1. $PC' = PC + 1$ for non-control-flow instructions; JUMP, JNEZ, CALL, and RET set PC to the target address; HALT stops execution.
2. $\mu' = \mu + \Delta\mu_i$.
3. $err' = err \vee \text{fail}(S, i)$.

5.2 Traces and Execution

Definition 5.1 (Trace). A trace $\tau = [i_0, \dots, i_k] \in \mathcal{I}^*$ is a finite sequence of instructions.

Definition 5.2 (Execution). $\text{Run}([], S) = S$; $\text{Run}(i :: \tau', S) = \text{Run}(\tau', \delta(S, i))$.

Theorem 5.1 (Unique descended values, **(S)**). *Let $V(t)$ be evaluation of a VM instruction list from `init_state`, and let $T(t)$ be target evaluation from a fixed basepoint. Then*

$$(\forall t, u, V(t) = V(u) \Rightarrow T(t) = T(u))$$

*is equivalent to each reachable VM state having exactly one target value represented by its traces. This is **trace_descent_unique_value_iff**.*

Theorem 5.2 (Constructive reachable simulation, **(S)**). *Supply a function r selecting an instruction list for each VM state, satisfying $V(r(V(t))) = V(t)$ for every trace t . A map preserving the basepoint, steps on reachable states, and certification exists exactly when target evaluation respects equal VM outcomes and target certification agrees with VM certification on every trace. The map is $s \mapsto T(r(s))$ and is pointwise unique on reachable states. This is **reachable_simulation_exists_iff** together with **reachable_simulation_unique**. No selector construction or extension of the step law to unreachable states is asserted.*

The certification quotient gives an explicit inhabited target. Conversely, the trace-history target satisfies A2 and certification agreement but fails descent: the empty trace and `JUMP 0 0` reach the same VM state while leaving distinct histories.

5.3 Selected Instruction Semantics

PNEW: Normalize region; if already present, idempotent; else add $(R_{\text{norm}}, \emptyset)$ to G .

REVEAL: Update CSRs with checksum of certificate cert . Cost: $\text{bits} + S(\Delta\mu)$.

LASSERT: Read formula from $\text{vm_mem}[\text{R}[\text{formula_addr_reg}]]$ (encoded-unit count formula_len , eight concrete bits per unit) and witness package from $\text{vm_mem}[\text{R}[\text{cert_addr_reg}]]$. $\text{cert_kind} = \text{true}$ (SAT): verify using a satisfying model plus a falsifying countermodel (dual-witness). $\text{cert_kind} = \text{false}$ (UNSAT): the kernel step always fails at this path, lassert_check_ok returns **false** unconditionally; LRAT checking is not invoked in the step semantics (documented gap in `VMStep.v`). If valid, the constraint is certified; if invalid, set $\text{err} \leftarrow \text{true}$. Note: in the current hardware-aligned semantics, LASSERT validates the constraint but does not directly mutate the partition graph's axiom set, cert-channel activation happens through the subsequent CERTIFY or MORPH_ASSERT. Cost: $\text{formula_len} \times 8 + S(\mu_delta)$.

Chapter 6

No Free Insight

The central theorem: structural insight is never free.

6.1 Definitions

Definition 6.1 (Receipt Predicate). $P : \mathcal{O} \rightarrow \mathbb{B}$ is a computable function on observation histories.

Definition 6.2 (Strength Ordering). $P_1 \leq P_2$ iff $\forall o. P_1(o) = 1 \implies P_2(o) = 1$. $P_1 < P_2$ iff $P_1 \leq P_2$ and $\exists o. P_2(o) = 1 \wedge P_1(o) = 0$.

Definition 6.3 (Certification). Trace τ **certifies** P iff execution succeeds ($err = 0$), the supra-certificate flag is set, and the output satisfies P .

Definition 6.4 (Structure Addition). $\text{HasStructureAddition}(\tau)$ is true iff τ contains a **REVEAL**, **LASSERT**, or equivalent operation.

6.2 The Theorem

Theorem 6.1 (No Free Insight (`NoFreeInsight_Theorem.v`)). *Let $P_{\text{strong}} < P_{\text{weak}}$. If a trace τ starting from a clean state s_0 certifies P_{strong} , then:*

$$\text{Certified}(\tau, P_{\text{strong}}) \implies \text{HasStructureAddition}(\tau)$$

Proof Sketch. The proof is parametric: `NoFreeInsight_Theorem.v` proves the theorem for any system satisfying the `NO_FREE_INSIGHT_SYSTEM` module type interface. The interface requires four contract obligations (not Coq `Axiom` declarations, they are proved separately for the kernel instantiation): (1) μ -monotonicity, (2) a certification specification linking **certifies** to execution, (3) a no-free-insight contract stating that strict strengthening requires structure addition, and (4) that the system has a notion of clean start. The theorem file itself contains zero `Axiom`, zero `Admitted`. The contract obligations are discharged by the kernel instantiation. $\square$

Corollary 6.2 (Cost of Insight). *Gaining insight (strengthening the predicate) implies $\Delta\mu > 0$.*

6.3 μ -Ledger Necessity

Theorem 6.3 (Observation adequacy, (**S**)). *For observation $o : S \rightarrow V$ and query $q : S \rightarrow A$, the existence of a decoder $d : V \rightarrow A$ satisfying $d(o(s)) = q(s)$ implies*

$$o(s) = o(t) \implies q(s) = q(t).$$

Given a section $r : V \rightarrow S$ with $o(r(v)) = v$, this condition is also sufficient: $d = q \circ r$. These are `decoding_requires_fiber_constancy` and `selected_representatives_give_decoder`.

Theorem 6.4 (μ -Ledger Necessity and Complete Independence Classification (NecessityOfMuLedger.v, kernel/NecessityAbstract.v), (S)). *Five independence results are formally proven, zero Admitted:*

1. $\mu \perp (\text{mem}, \text{regs}, \text{pc})$: no strict-classical oracle recovers μ *turing_ram_mu_necessity*
2. $\text{certified} \perp (\text{mem}, \text{regs}, \text{pc})$: no strict-classical oracle recovers certification *turing_ram_cert_necessity*
3. $\text{certified} \perp (\text{mem}, \text{regs}, \text{pc}, \mu)$: knowing μ does not determine certification *cost_model_cert_necessity*
4. $\mu \perp (\text{mem}, \text{regs}, \text{pc}, \text{certified})$: knowing certification does not determine μ *cert_model_mu_necessity*
5. $\text{vm_graph} \perp (\text{mem}, \text{regs}, \text{pc}, \mu, \text{certified})$: the categorical layer is independent of the full μ -ledger shadow *graph_not_recoverable_from_P_full*

Three-component independence (*thiele_state_three_component_independence*): μ , vm_certified , and vm_graph are the three provably independent non-classical state components. Each is irrecoverable from any combination of the others plus $(\text{mem}, \text{regs}, \text{pc})$.

Minimality (*P_full_complete_neither_mu_nor_cert_droppable*): $(\text{mem}, \text{regs}, \text{pc}, \mu, \text{certified})$ is the minimal complete extension of classical state for μ and certification. Graph structure is a third independent axis.

Universality (*mu_ledger_necessity_universal*): The separation holds from every VM-State, not only from crafted zero-state witnesses. From any reachable state, **CERTIFY** 0 and **PNEW** [] 0 produce identical strict classical shadows but different μ . Certification also differs when the source state is uncertified.

Trace closure (*po1_trace_necessity*): For any common prefix program of any length, the two witness extensions maintain equal strict shadows but strictly different μ . No accumulation of classical computation closes the gap.

Proof Sketch. Each separation uses explicit witnesses where equal projected inputs force equal oracle outputs, while the witness states require unequal outputs, a contradiction. Strict shadow equality: both **CERTIFY** 0 and **PNEW** [] 0 advance pc by 1 and preserve mem/regs . μ divergence: **CERTIFY** charges $S(0) = 1$; **PNEW** charges 0. Graph separation uses **categorical_separation** from **PartitionSeparation.v**: two states with identical P_{full} but different pg_morphisms . Universality follows because **vm_apply_certify_strict_shadow** and **vm_apply_pnew_strict_shadow** hold for arbitrary source states. $\square$

Theorem 6.5 (Latent μ : Trace Cost for a Fixed Schedule (NecessityOfMuLedger.v §7), (S)). *The μ -cost accumulated by any computation is a property of the instruction sequence, not of any tracking system. Four results, all zero Admitted:*

1. **shadow_mu_is_computation_intrinsic**: For any program prog and any starting state s :

$$(\text{run_instrs } s \text{ prog}).\mu = s.\mu + \text{trace_total_cost}(\text{prog})$$

The increment is determined entirely by the instruction sequence, independent of vm_graph , vm_regs , vm_mem , vm_certified , and the starting μ .

2. **shadow_mu_delta_universal**: Two machines running the same program from any two starting states accumulate the same additional μ . The cost is written in the program before execution begins.

3. *shadow_mu_unique_accounting*: Any instruction-consistent accounting system M satisfying $M(s) = 0$ at the starting state assigns exactly `trace_total_cost` to the result. Replace the Thiele Machine's μ -counter with any other honest accounting and you get the same number.
4. *shadow_mu_inevitable*: Any program containing an instruction with positive declared cost accumulates strictly positive μ from any starting state. This conclusion requires a positive-cost instruction.

Interpretation. Given a trace and this cost schedule, its sum is determined even if the observer does not retain the ledger. This does not establish a physical bill for every classical computation. A full encoding may retain the ledger; the irrecoverability theorem concerns the specified projection.

6.4 Partition Refinement and No Free Insight

Theorem 6.6 (Partition Ops Are Free (kernel/PartitionRefinementNoFI.v), (S)). *PNEW, PSPLIT, and PMERGE are not cert-setters and can carry $\Delta\mu = 0$:*

- *partition_structural_ops_not_cert_setters*: None of the three partition operations touch `csr_cert_addr` or `vm_certified`.
- *partition_structural_ops_can_be_free*: All three can execute with $\Delta\mu = 0$.
- *partition_structural_trace_cannot_certify*: A trace containing only PNEW, PSPLIT, and PMERGE (all at $\Delta\mu = 0$) cannot reach a certified state from an uncertified start.

Building partition structure costs nothing by itself. Certifying structural claims about that partition is what costs.

Theorem 6.7 (Partition Free But Certification Non-Free (kernel/PartitionRefinementNoFI.v), (S)). *partition_free_but_certification_nonfree*: There exist traces where all partition-structural instructions carry $\Delta\mu = 0$ (so $\mu_{\text{final}} = \mu_{\text{init}} = 0$) yet the trace ends with a partition graph that a subsequent MORPH_ASSERT or CERTIFY can certify, but only at $\Delta\mu \geq 1$. Partition structure can be built at zero μ -cost; certified partition knowledge cannot be acquired for free. Zero Admitted.

Theorem 6.8 (Partition Refinement Non-Free (kernel/PartitionRefinementNoFI.v), (S)). *partition_refinement_nonfree*: Any trace that starts with no cert evidence and ends with a certified claim about a partition structure paid at least 1 in μ . This is the NoFI law specialized to partition refinement. Zero Admitted.

Chapter 7

The μ -Chaitin Theorem

A Chaitin-style incompleteness theorem denominated in μ -currency.

Theorem 7.1 (μ -Chaitin (MuChaitinTheory_Theorem.v)). *For any formal theory system T satisfying the μ -Chaitin interface, the number of bits k the theory can certify is bounded by:*

$$k \leq |T| + c$$

where $|T|$ is the description length of the theory and c is a fixed overhead constant. The proof chain: $k \leq \text{payload} \leq \mu\text{-info} \leq \text{budget} \leq |T| + c$.

Chapter 8

Computational Universality and Subsumption

8.1 Strict Containment

Theorem 8.1 (Turing $\subsetneq$ Thiele, as instruction sets (kernel/Subsumption.v)). *Classical Turing computation is **properly extended** by sighted Thiele computation (both are Turing-complete, but Thiele carries additional partition and μ -accounting structure):*

$$\exists p. \text{sighted}(p) \wedge \neg \text{turing}(p)$$

There exist programs that are Thiele programs (using partition structure and μ -accounting) but are not classical Turing programs. The witness is the one-instruction program `[H_ClaimTapeIsZero 0]`. The separation is syntactic: the instruction language is larger. It is not a claim that the Thiele model computes any function a Turing machine cannot. The extra structure is new content in what a program can say, not in what it can compute. The embedding in the other direction lives in `TuringClassicalEmbedding.v`.

8.2 Oracle Cost Lower Bound

Theorem 8.2 (Oracle pricing self-consistency (conceptual; not separately formalized)). *Inside the No Free Insight framework, the universal lower bound on certification cost (`universal_nfi_any_substrate` in `coq/kernel/nfi/UniversalCertificationCost.v`) immediately implies that any sound oracle satisfying the one-step rule `cs_cert_costs` ($\text{cost} \geq 1$ on any uncertified-to-certified transition) charges $\geq n$ for n certifying queries. There is no separate oracle-impossibility file in the codebase; the oracle case is a direct instantiation of the universal theorem at a `CertificationSystem` whose state encodes oracle responses. The justification for charging $\mu > 0$ on oracle queries comes from NoFI, which is an accounting law, not a physical one: gaining certified information about undecidable propositions has positive μ -cost.*

8.3 μ -budget predicates

Definition 8.1 (μ -budget predicates (kernel/MuComplexity.v)). Let $p_{\text{fuel}}, p_{\mu} : \mathbb{N} \rightarrow \mathbb{N}$ be numeric bounds and $\text{size} : \text{VMState} \rightarrow \mathbb{N}$ a size function. Polynomiality is not required.

- `error_free_preservation_budget` means that every input has an error-free trace within the length and μ bounds, and that f has equal values on the initial and final states whenever μ increases.

- `positive_input_certification_budget` means that every input satisfying f has a trace within those bounds ending with `vm_certified = true`.

Theorem 8.3 (Zero- μ traces satisfy the preservation budget (kernel/MuComplexity.v), (S)). *zero_mu_traces_satisfy_preservation_budget: if every input has a bounded, error-free, zero- μ trace, the preservation budget holds with μ -bound zero for every Boolean state predicate. Its positive- μ antecedent never holds.*

Remark 8.1. These predicates supply neither a uniform solver with a correct output nor a sound verifier rejecting negative inputs. They do not define decision or verification complexity classes. The implication between them is named as a premise, not proved. The factored-search witness in `StructuralAdvantage.v` gives a concrete program-level μ -cost advantage (paying $\mu = 18$ to certify a decomposition and reduce search from $O(N^2)$ to $O(2N)$), but that is not a class separation.

Theorem 8.4 (μ -Cost Hierarchy (kernel/MuHierarchyTheorem.v), (S)). *For every $k \geq 1$, the μ -dimension admits a strict hierarchy:*

1. **Achievability:** *there exists a trace from `init_state` with μ -cost exactly k that is level- k certified, the run executes a `CERTIFY` δ instruction with $S(\delta) \geq k$ and ends with `vm_certified = true`.*
2. **Lower bound:** *any level- k certified trace from `init_state` must have spent at least k units of μ -cost.*

Formally (*mu_hierarchy_theorem*): for every $k \geq 1$, the witness trace `[instr_certify(k-1)]` ($\text{cost} = S(k-1) = k$) executes a level- k `CERTIFY` from `init_state`; conversely, the trace-side predicate `level_k_certified` forces $\text{trace_mu_cost} \geq k$ via `level_k_certification_cost_floor`, since the `CERTIFY`'s declared cost appears in the executed-instruction ledger and the ledger sum equals the trace cost (*run_vm_mu_conservation*).

Corollary (*mu_hierarchy_no_upper_bound*): *no fixed μ -budget suffices for all levels. The μ -dimension is unbounded and non-collapsible. Zero Admitted.*

Remark 8.2 (μ -Hierarchy as a P-vs-NP analogue, not a P-vs-NP result). Theorem 8.4 is the μ -analogue of the classical time hierarchy theorem in shape. Its content is lighter. The level of a certificate is defined by the operand δ of the `CERTIFY` that issued it, and the cost schedule charges $\delta + 1$ for that instruction, so the floor comes from the schedule, not from a search or diagonal argument. It is not a statement about classical P versus NP, which this project does not touch. In the certified-structure setting it makes one thing precise: issuing a certificate at level k costs $\mu \geq k$, because the schedule charges $\delta + 1$ for the `CERTIFY` that sets the level. It says nothing about how hard the certified thing was to find. The hierarchy is strict and unbounded, with zero Admitted, no oracle, and no diagonalization. The lower-bound proof is trace-side: a level- k certified run executed a specific `CERTIFY` δ with $S(\delta) \geq k$, and `executed_instruction_cost_recorded` plus `ledger_sum_contains_lower_bound` place that cost in the ledger sum, which `run_vm_mu_conservation` equates with the trace's μ -cost.

8.4 Intrinsic level hierarchy

Definition 8.2 (Intrinsic level (coq/IntrinsicLevelHierarchy.v)). A final state s is *intrinsically certifiable at level $\geq k$* , written `level_intrinsic_at_least(k, s)`, iff every `run_vm` trace from `init_state` that ends at s with `vm_certified = true` executes at least k cert-setter instructions:

$$\begin{aligned} & \forall \text{fuel trace. run_vm}(\text{fuel}, \text{trace}, \text{init_state}) = s \\ & \wedge s.\text{vm_certified} = \text{true} \Rightarrow \text{cert_setter_executions}(\text{fuel}, \text{trace}, \text{init_state}) \geq k. \end{aligned}$$

The level is a property of the *final state*, not of any particular trace reaching it.

Theorem 8.5 (Intrinsic level forces μ (coq/IntrinsicLevelHierarchy.v::level_intrinsic_forces_mu)). *If $\text{level_intrinsic_at_least}(k, s)$ holds and $s.\text{vm_certified} = \text{true}$, then for every trace reaching s from init_state , $s.\text{vm_mu} \geq k$. Zero Admitted.*

Theorem 8.6 (Strict separation (coq/IntrinsicLevelHierarchy.v::level_k_plus_1_forces_more_than_k_cert_events)). *For every $k \geq 0$, if $\text{level_intrinsic_at_least}(k + 1, s)$ holds and $s.\text{vm_certified} = \text{true}$, then every trace reaching s executes strictly more than k cert-setter instructions. Zero Admitted.*

Remark 8.3 (Relation to Theorem 8.4). Theorem 8.6 is the state-level counterpart of the trace-level μ -hierarchy. The trace-level version (`mu_hierarchy_theorem`, `level_k_certification_cost_floor`) defines level on the trace, a trace is level- k certified iff it executes a `CERTIFY` δ with $S(\delta) \geq k$. The state-level version defines level on the final state, every trace reaching it requires at least k cert-setter executions. The two are complementary entry points to the same hierarchy: the trace-level theorem gives the achievability witness (`[instr_certify(k-1)]` from `init_state`); the state-level theorem gives the lower bound on μ for any state intrinsically requiring k cert-events. Both are present-tense in the kernel; both close under the global context.

8.5 Verifier model and the bare-setting impossibility

Definition 8.3 (Bare verifier (coq/VerifierModel.v)). The bare-setting transcript type is `BareTranscript := list StrictClassicalState`. A bare verification problem is a record `BareVerificationProblem` carrying

- `bare_honest_claim : VMState → Prop`;
- `bare_explains : VMState → BareTranscript → Prop` (the transcript-honestly-witnessed-by-state relation);
- `bare_cheap_budget, bare_recompute_cost : ℕ`.

A bare verifier is a record `BareVerifier` with `bv_decide : BareTranscript → bool` and `bv_cost : BareTranscript → ℕ`.

Definition 8.4 (Soundness, completeness, cheapness (coq/VerifierModel.v)). For a problem P and verifier V :

- $\text{bare_sound}(P, V) := \forall t. V(t) = \text{true} \Rightarrow \forall s. \text{bare_explains}(P, s, t) \Rightarrow \text{bare_honest_claim}(P, s)$.
- $\text{bare_complete}(P, V) := \forall s t. \text{bare_honest_claim}(P, s) \Rightarrow \text{bare_explains}(P, s, t) \Rightarrow V(t) = \text{true}$.
- $\text{bare_cheap}(P, V) := \forall t. \text{bv_cost}(V, t) \leq \text{bare_cheap_budget}(P)$.

The soundness condition is *strong*: the claim must hold for every state explaining the transcript, not merely for some.

Theorem 8.7 (Bare-setting impossibility (coq/VerifierImpossibility.v::bare_setting_no_sound_complete_verifier)). *There is no bare verifier simultaneously `bare_sound` and `bare_complete` on the verification problem `mu_eq_one_problem` whose claim is $\text{vm_mu}(s) = 1$ and whose `bare_explains` relation holds for the two witness states from `coq/NecessityOfMuLedger.v` against the shared strict-shadow trace.*

Proof. `po1_state_A` (the cost-1 CERTIFY 0 result, $\mu = 1$) and `po1_state_B` (the cost-0 PNEW [] 0 result, $\mu = 0$) project to identical strict-shadow traces. Completeness applied to A 's honest run forces V to accept the shared transcript. Strong soundness applied to the same accepted transcript at the explanation by B forces $\text{vm_mu}(B) = 1$, contradicting $\text{vm_mu}(B) = 0$. `Print Assumptions` returns Closed under the global context. $\square$

Theorem 8.8 (Three sufficient interfaces (`coq/VerifierEscape_{Substrate,Hardness,Interaction}.v`)). *Each kernel theorem below exhibits a constant-cost verifier that simultaneously satisfies soundness and completeness on the μ -sensitive claim by augmenting the transcript with non-classical structure:*

- *`substrate_escape_succeeds`, transcript is `list VMState`; verifier reads μ directly.*
- *`commitment_contract_verifier`, transcript is `BareTranscript` $\times$ `bool` (commitment bit); the `CommitmentBitContract` supplies exact agreement between the bit and the claim for every explaining state. The verifier reads the bit at abstract unit cost. No cryptographic security or runtime bound is proved.*
- *`interactive_escape_succeeds`, transcript is `BareTranscript` $\times$ $\mathbb{N}$ (response); verifier checks the response against the claim.*

Each closes under the global context with zero Admitted.

Theorem 8.9 (Factorisation impossibility (`coq/VerifierExhaustiveness.v::V_does_not_factor_through_classical`)). *Let T be a transcript type, $\pi : T \rightarrow \text{BareTranscript}$ a projection, and explains_T an explanation relation. Given supplied colliding transcripts $t_A, t_B : T$ such that $\pi(t_A) = \pi(t_B)$, where t_A explains the fixed $\mu = 1$ witness and t_B explains the fixed $\mu = 0$ witness, any verifier sound and complete for the named claim satisfies $\neg \text{factors_classical } \pi V$.*

Remark 8.4 (The verifier corollary). The collision and both explanation premises are hypotheses, not conclusions. The theorem does not construct a collision for arbitrary T or π , and it does not classify all verifier designs. The full-state transcript, exact commitment-bit contract, and reported response above are three sufficient interfaces. The commitment-bit theorem is not a computational-hardness result. The substrate channel is the option the structural axis makes available; hardness and interaction are the two routes classical cryptography and complexity already used, and here the hardness route appears as the exact commitment-bit contract.

Chapter 9

μ -Conservation and Receipt Integrity

9.1 The μ -Ledger Identity

Theorem 9.1 (μ -Ledger Conservation (MuLedgerConservation.v)). *For any trace $\tau = [i_0, \dots, i_k]$:*

$$\mu_{\text{final}} = \mu_{\text{init}} + \sum_{j=0}^k \text{cost}(i_j)$$

Every instruction increases μ by exactly its declared cost. No axioms, no admits.

Corollary 9.2 (Monotonicity). *μ never decreases: $\mu(s) \leq \mu(\text{Run}(\tau, s))$ for all τ .*

Corollary 9.3 (Irreversibility Bound). *$\text{irreversible_bits}(\tau) \leq \mu_{\text{final}} - \mu_{\text{init}}$.*

Theorem 9.4 (μ Decomposition (MuLedgerConservation.v)). *$\mu_{\text{total}} = \mu_{\text{blind}} + \mu_{\text{sighted}}$ for any partition of the cost into blind (reversible) and sighted (structural) components.*

9.2 Receipt Integrity

Definition 9.1 (Receipt). A receipt $r = (\text{step}, \text{instr}, \mu_{\text{pre}}, \mu_{\text{post}}, h_{\text{pre}}, h_{\text{post}})$ records one transition.

Theorem 9.5 (Receipt Integrity (ReceiptIntegrity.v)). *A valid receipt or execution attestation proves $\mu_{\text{final}} = \mu_{\text{init}} + \sum \text{costs}$ for the attested run. Any receipt claiming $\Delta\mu \neq \text{cost}(\text{instr})$ fails validation.*

Theorem 9.6 (Non-Forgeability (ReceiptIntegrity.v)). *Forged receipts (claiming incorrect $\Delta\mu$) fail validation. Overflow values ($\mu > 2^{31} - 1$) are rejected at the receipt layer.*

Chapter 10

Gauge Shifts of the Ledger and a Noether-Style Invariance

10.1 The Partial $\mathbb{Z}$ -Action (Gauge Group)

Definition 10.1 (Gauge Shift). For $\delta \in \mathbb{Z}$, the gauge shift $\sigma_\delta : \mathcal{S} \rightarrow \mathcal{S}$ maps $S \mapsto S[\mu \leftarrow \text{toNat}(\mu + \delta)]$, preserving all other fields.

Note the truncation. Since μ ranges over $\mathbb{N}$, the Coq definition `z_gauge_shift` computes `Z.to_nat (Z.of_nat vm_mu + delta)`, and `Z.to_nat` clamps negatives to zero. So σ_δ is a total function on states but is *not* injective, and the shifts do not form a $\mathbb{Z}$ -action on all of $\mathcal{S}$.

Theorem 10.1 (Partial Group Action (KernelNoether.v)). *The gauge shifts form a partial $\mathbb{Z}$ -action on states; equivalently, a genuine $\mathbb{Z}$ -action on the sub-poset of states where no intermediate shift underflows:*

1. **Identity:** $\sigma_0(S) = S$. Unconditional (`z_action_identity`).
2. **Composition:** $\sigma_a(\sigma_b(S)) = \sigma_{a+b}(S)$ provided $\mu(S) + b \geq 0$ and $\mu(S) + b + a \geq 0$ (`z_action_composition`, which carries both hypotheses).
3. **Inverse:** $\sigma_{-n}(\sigma_n(S)) = S$ provided $\mu(S) + n \geq 0$ (`z_action_inverse`).

Remark 10.1 (Why composition needs its guard). The guard on clause 2 is not decoration. Take $\mu(S) = 0$, $b = -1$, $a = +1$: σ_b clamps to $\mu = 0$, so $\sigma_a(\sigma_b(S))$ has $\mu = 1$; whereas $\sigma_{a+b}(S) = \sigma_0(S) = S$ has $\mu = 0$. The two sides differ. Composition fails wherever an intermediate shift underflows, which is why the mechanised theorem carries *two* nonnegativity hypotheses: one for the intermediate state, one for the final state. Drop either and the counterexample above goes through.

10.2 Gauge Invariance

Definition 10.2 (Observable). $\text{Obs}(S)$ extracts the partition region structure, ignoring μ .

Theorem 10.2 (Gauge Invariance (KernelNoether.v)). $\text{Obs}(\sigma_\delta(S)) = \text{Obs}(S)$. *Shifting μ by a constant does not affect observables.*

Theorem 10.3 (Noether-Style Orbit Invariance (`coq/kernel/curvature/KernelNoether.v`)).

1. **Forward:** *States identical except for μ lie on the same gauge orbit. (This is direct from the definition: if all other fields match, the μ difference is the gauge shift.)*

2. **Gauge Invariance:** `z_gauge_invariance` proves that shifting μ does not change `Observable_partition`. This is definitional: `z_gauge_shift` only modifies the `vm_mu` field, and `Observable_partition` reads only `vm_graph`.

3. **Trace Preservation:** Gauge-equivalent states produce identical observable traces (same labels, same μ -costs) for any finite horizon. Proven as `vm_step_orbit_equiv`.

The μ -ledger is a gauge degree of freedom. Its absolute value is unobservable; only differences $\Delta\mu$ carry meaning. The “Noether” label is a deliberate analogy: the result follows from the record structure (separate fields for μ and `graph`), not from a deep symmetry principle.

Remark 10.2 (Gauge freedom and absolute μ -claims). Gauge freedom has to be squared with the results that assert absolute values, most directly the verifier argument, which needs `vm_mu(s) = 1` to be a well-posed claim, and the μ -hierarchy, which needs “cost exactly k ”. If only differences carried meaning, neither statement would have content.

Every absolute μ -claim in this document is *relative to `init_state`*, where $\mu = 0$ by definition. Read `vm_mu(s) = 1` as $\Delta\mu = 1$ along a run from `init_state` to s ; μ -conservation (`run_vm_mu_conservation`) is what licenses the reading, since it makes the final ledger equal to the trace cost.

The scope is worth naming precisely: σ_δ is defined on *arbitrary* states, not on orbits rooted at `init_state`, so the gauge group as formalised is larger than the equivalence the absolute claims need. Nothing in the development quotients by it. The honest statement is that the theorems live on *reachable* states, where the base point pins the gauge, and the gauge chapter describes a symmetry of the state *type* rather than of the reachable dynamics. Note also that `vm_step_mu_monotonic` makes μ non-decreasing along runs, which already breaks the $\mathbb{Z}$ -symmetry on trajectories: the dynamics single out a direction the gauge action does not respect.

Theorem 10.4 (μ -Monotonicity (KernelNoether.v)). $\text{vm_step}(S, i, S') \implies \mu(S) \leq \mu(S')$. The μ -ledger never decreases under the dynamics. This is true by construction: instruction costs are natural numbers (≥ 0), so adding them to the ledger cannot decrease it. The proof examines each step constructor and confirms the arithmetic.

Remark 10.3 (Part V Scope: Conditional Results and Named Axioms). The chapters in this part present results conditional on named physical axioms (linearity in the Bloch parameter for the Born rule, the second law for Landauer, and the `A_QM` bridge from NPA PSD to quantum realizability). They are marked **(C)** or **(R)** throughout. The primary unconditional quantum results, the classical CHSH bound, the algebraic Tsirelson bound, and the biconditional `column_contractive_iff_npa_psd`, appear in the chapters that follow the conditional ones: the classical CHSH bound in Chapter 17, the algebraic Tsirelson bound in Chapter 15, and the biconditional in Chapter 18. Those are pure algebra, no physics axioms.

The results in this part are not primary claims of the project. They are explorations of what follows *if* you add specific physical axioms. They show how far the machine reaches without claiming any physics the proofs do not support.

Chapter 11

The Born Rule

Definition 11.1 (Bloch Sphere Probabilities). For a state with purity vector (x, y, z) :

$$P(|0\rangle) = \frac{1+z}{2}, \quad P(|1\rangle) = \frac{1-z}{2}$$

Definition 11.2 (Measurement μ -Cost).

$$\text{Cost}(x, y, z) = \frac{1 - (x^2 + y^2 + z^2)}{2}$$

This is the linear entropy. For pure states ($r^2 = 1$), $\text{cost} = 0$. For mixed states ($r^2 < 1$), $\text{cost} > 0$.

Theorem 11.1 (Uniqueness of the Born Rule among Linear Rules (BornRule.v), (C)). *The Born rule $P(|0\rangle) = (1+z)/2$ is the **unique** probability assignment satisfying:*

1. *Non-negativity and normalization:* $P \geq 0$, $P(|0\rangle) + P(|1\rangle) = 1$.
2. *Eigenstate consistency:* $P(0, 0, 1) = 1$, $P(0, 0, -1) = 0$.
3. *Linearity in z :* $P(x, y, z) = az + b$ for some a, b .

Proof: Boundary conditions force $a = 1/2$, $b = 1/2$. No other solution exists.

Note: The Coq theorem also lists a `receipt_mu_consistent` hypothesis, but it is **unused** in the proof. The hypothesis `receipt_mu_consistent` P unfolds to “measurement cost ≥ 0 for valid Bloch vectors,” which is directly from the definition true for all states and does not constrain P . The linearity assumption (item 3) does all the work, it assumes the functional form $P = az + b$ and solves a 2-equation system. This is less “deriving the Born rule from μ -accounting” and more “the Born rule is the unique linear rule consistent with eigenstates.”

Chapter 12

Unitarity

Definition 12.1 (Evolution). An evolution E maps Bloch vectors $(x, y, z) \mapsto (x', y', z')$ with associated μ -cost. The cost is a real field of the record. Nothing in the file computes it from the map.

The predicate **radius_preserving** means preservation of squared Bloch radius on the unit ball. The file does not prove that such a map comes from a Hilbert-space operator. Two premises are supplied, not derived: **respects_info_conservation** (radius loss is at most μ) and **purity_nonincreasing** (radius gain is at most μ).

Theorem 12.1 (Zero Cost and Radius Preservation (Unitarity.v), (C)).

1. Under the conservation premise, $\mu = 0 \implies r'^2 \geq r^2$ (purity cannot decrease at zero cost). Proven as **zero_cost_preserves_purity**.
2. Under the conservation premise, non-unitary evolution ($r'^2 < r^2$ for some state) requires $\mu > 0$. Proven as **nonunitary_requires_mu**: the proof specializes the conservation hypothesis at the witness, then **lra**.
3. Under both premises, $\mu = 0$ implies **radius_preserving**. Proven as **zero_cost_preserves_radius**: the two premises give $r'^2 \geq r^2$ and $r'^2 \leq r^2$.
4. The converse, radius preservation $\implies \mu = 0$, is stated in the file but not proven as a theorem. It appears as the **Definition radius_preserving_zero_cost**. Because μ is a free field of the record, it could not be proved for every evolution.

Theorem 12.2 (A Dissipation Witness Forces $\mu \geq \gamma$ (Unitarity.v), (C)). Assume the conservation premise, a supplied bound **satisfies_lindblad_bound** $E \gamma$ with $\gamma > 0$, and a witness that the pure state $(1, 0, 0)$ loses exactly γ of squared radius. Then $\mu \geq \gamma$ (**lindblad_requires_mu**). In this record, that loss is paid for. The file names the bound after Lindblad dynamics, but it constructs no Lindblad equation. Its “dissipation rate” is the radius loss, with no time derivative.

Theorem 12.3 (The Scalar Ball-and-Trace Contract (Unitarity.v), (R)). Positivity and trace preservation imply **ball_and_trace_preserving** (**ball_trace_contract_from_premises**). The theorem packages the supplied unit-ball and trace conditions. It has no ancillary-system test and establishes no complete positivity in the operator sense.

Chapter 13

Purification

Theorem 13.1 (Two-Point Decomposition of a Bloch State (Purification.v), (S)). *The file calls this the purification principle. It proves the spectral fact that a purification would use. It does not construct a purifying state on a larger system. Every mixed state (x, y, z) with $r^2 = x^2 + y^2 + z^2 \leq 1$ admits eigenvalues λ_1, λ_2 with:*

- $\lambda_1 + \lambda_2 = 1, \quad 0 \leq \lambda_i \leq 1.$
- $(\lambda_1 - \lambda_2)^2 = r^2.$
- *Construction:* $\lambda_1 = (1 + r)/2, \lambda_2 = (1 - r)/2.$

Pure states ($r = 1$) need no external reference (deficit = 0). The maximally mixed state ($r = 0$) has deficit = 1.

Chapter 14

No-Cloning

A cloning operation here is a record of four real numbers: input information I , output informations I_1, I_2 , and a cost μ . The chapter's results are arithmetic on those fields under one supplied premise. They use no Hilbert space, channel, or fidelity between states.

Theorem 14.1 (No Perfect Cloning at Zero Declared Cost (NoCloning.v), **(C)**). *Let $I = x^2 + y^2 + z^2$ be the information content (purity). For a cloning operation with outputs of information I_1, I_2 and μ -cost μ , assume the conservation premise (**respects_conservation**, supplied, not derived):*

$$I_1 + I_2 \leq I + \mu$$

*If $I > 0$ and $I_1 = I_2 = I$, then $\mu \neq 0$ (**no_cloning_from_conservation**). For a pure input, $\mu \geq 1$ (**no_cloning_bloch**). **unitary_cannot_clone** links this to the Unitarity chapter. Under that chapter's two premises and $\mu = 0$, one evolution's output copied into both fields cannot satisfy the premise.*

Theorem 14.2 (Approximate Cloning (NoCloning.v), **(C)**). *Under the same premise, with $I > 0$ and $I_i = f_i I$ for scale factors $0 \leq f_i \leq 1$: $f_1 + f_2 \leq 1 + \mu/I$. At zero cost: $f_1 + f_2 \leq 1$. The file calls f_1, f_2 fidelities. They are scale factors on the information field, not the fidelity between quantum states.*

Theorem 14.3 (No Deletion at Zero Declared Cost (NoCloning.v), **(C)**). *Perfect deletion also requires $\mu \geq I$ (dual of no-cloning), under the deletion record's own supplied premise $I_{\text{out}} + \mu \geq I_1 + I_2$, with $I_{\text{out}} = I_1 = I_2 > 0$ (**no_deletion_without_cost**).*

Chapter 15

The Tsirelson Bound

Remark 15.1 (Epistemological Status: **(S)** algebraic, no physics axioms). The upper bound appears in `coq/kernel/quantum/TsirelsonGeneral.v` (`tsirelson_from_row_bounds`). `AlgebraicCoherence.v` supplies a rational coherent witness with $S \geq 28284/10000$ (`tsirelson_rational_lower_witness`). That witness does not attain the irrational optimum. `TsirelsonGeneral.v` derives $S^2 \leq 8$ from pure algebra: a sum-of-squares identity proves each CHSH row contributes $\leq 2(e_1^2 + e_2^2)$, then Cauchy–Schwarz bounds the sum by $4 \sum e_i^2 \leq 8$. `TsirelsonFromAlgebra.v` provides a self-contained algebraic derivation with achievability witness ($e = \pm 1/\sqrt{2}$, giving $S = 2\sqrt{2}$) and a verified rational bound ($\sqrt{8} < 5657/2000$). Zero admits in both files.

Theorem 15.1 (Tsirelson Upper Bound for the Zero-Marginal NPA Model (`TsirelsonQuantumModel.v`, `TsirelsonFromAlgebra.v`), **(S)**). *If a trace’s zero-marginal NPA matrix is PSD (equivalently, the trace is slice-coherent, i.e. column-contractive), then $|S_{\text{CHSH}}| \leq 2\sqrt{2}$.*

The algebraic derivation (`TsirelsonFromAlgebra.v`) proves $S^2 \leq 8$ for all algebraically coherent correlators satisfying the three column-contractivity conditions, using a sum-of-squares identity and psatz. No physics premises. Pure polynomial arithmetic.

Chapter 16

Witness Insight Taxonomy

Theorem 16.1 (CHSH Trial Is Tier 0 (kernel/WitnessInsightGeneral.v), (S)). *chsh_trial_is_tier0: The CHSH_TRIAL instruction never touches either certification channel:*

- *chsh_trial_not_cert_addr_setter: CHSH_TRIAL does not change csr_cert_addr.*
- *chsh_trial_preserves_vm_certified: CHSH_TRIAL does not change vm_certified.*

Recording CHSH trial outcomes activates no certification channel, so No Free Insight puts no floor under it. The counters go up; the cert channels are untouched. Its charge is its own μ -delta operand, which may be 0. Zero Admitted.

Theorem 16.2 (Three-Tier Observation Taxonomy (kernel/WitnessInsightGeneral.v), (S)). *witness_insight_complete_taxonomy is the conjunction of three proved facts, one per tier. The tiers are a reading aid. The theorem does not prove that every instruction falls into exactly one of them, and Tier 2 events are also certified events.*

Tier 0 *Raw observation (can be free): Records data without activating any certification channel. CHSH_TRIAL is the canonical example. Cost: $\Delta\mu = 0$ is possible; no cert channel is touched.*

Tier 1 *Certified structural insight (always costs ≥ 1): Any instruction that activates csr_cert_addr or vm_certified is in this tier. Proven: certified_witness_insight_nonfree shows μ increases by at least 1.*

Tier 2 *Certified non-local witness insight (always costs ≥ 1): A certified execution whose WitnessCount statistics show $S > 2$. nonlocal_witness_insight_nonfree: getting certified with non-local evidence requires $\mu \geq 1$. In the taxonomy theorem this is a trace statement: a run of the certification machine from an uncertified state to a certified state with a certified CHSH violation raises μ by at least 1.*

witness_insight_nonfree_general: For any instruction in Tier 1 or trace in Tier 2, μ increases by at least 1. Of the three tiers, only Tier 0 can be free. Zero Admitted.

Chapter 17

Bell's Theorem in Morphism Coupling Language

Definition 17.1 (Setting-Outcome Coupling (kernel/CHSHCouplingBridge.v)). The `chsh_setting_outcome_coupling` maps a `WitnessCount` record to a coupling (list of $(nat \times nat)$ pairs). Each pair records a (setting-pair-index, outcome-type) observation. A coupling is **separable** (functional) when it maps each setting to at most one outcome type; **non-separable** otherwise.

Theorem 17.1 (Locally Consistent Strategy Gives Separable Coupling (kernel/CHSHCouplingBridge.v), (S)). *locally_consistent_gives_separable_coupling: Any WitnessCount configuration produced by a locally deterministic strategy $(a_0, a_1, b_0, b_1) \in \{0, 1\}^4$ gives a separable (functional) coupling. Zero Admitted.*

Theorem 17.2 (Local Coupling Bound (kernel/CHSHCouplingBridge.v), (S)). *locally_consistent_classical_bound: Any locally factorizable morphism coupling satisfying `WCLocallyConsistent` has $|S| \leq 2$. Bell's theorem in the morphism coupling language: the separation between local and non-local correlations is not just a statistical fact but a structural property of the coupling predicate. Zero Admitted.*

Theorem 17.3 (CHSH Violation Rules Out Local Couplings (kernel/CHSHCouplingBridge.v), (S)). *chsh_violation_rules_out_locally_factorizable_coupling: If WitnessCount statistics satisfy $S > 2$, no locally factorizable morphism coupling is consistent with those counts. Contrapositively: non-local correlations cannot be produced by any local morphism coupling. The no-go is stated in the coupling language, not just the statistical layer. Zero Admitted.*

Remark 17.1. These results connect the categorical morphism layer directly to Bell nonlocality. A claim that two modules are entangled (via `MORPH_ASSERT` on a non-separable coupling) cannot be made for free: the NoFI law charges at least 1 for the `MORPH_ASSERT`, and the coupling theorem rules out any locally factorizable coupling consistent with counts showing $S > 2$. “Entangled” is the reading of a non-separable coupling. The theorems speak about counts and couplings, not about physical systems.

Chapter 18

Slice Coherence and NPA Realizability: The Biconditional

This algebraic result has no project-local axioms or admitted lemmas; its assumption receipt includes standard classical-real and functional-extensionality axioms.

Definition 18.1 (Column Contractivity (Slice Coherence)). **18.1 The CHSH statistic S and its witness counts**

The central quantum object of this part is defined here, since it is used throughout without ever having been given as a function of machine state.

Definition 18.2 (Correlator from witness counts (CHSHStatisticalBridge.v)). A VM state carries a **WitnessCounts** record in $\mathbb{N}^8$, holding for each setting pair $(x, y) \in \{0, 1\}^2$ the tallies `samexy` and `diffxy`. Writing $N_{xy} = \text{same}_{xy} + \text{diff}_{xy}$, the empirical correlator is

$$E_{xy} = \begin{cases} \frac{\text{same}_{xy} - \text{diff}_{xy}}{N_{xy}} & N_{xy} > 0, \\ 0 & N_{xy} = 0. \end{cases} \quad (18.1)$$

The $N_{xy} = 0$ branch is a definitional convention (no trials, no evidence), not a claim that uncorrelated is the right default.

Definition 18.3 (CHSH statistic (chsh_S, TsirelsonFromMu.v)).

$$S = E_{00} + E_{01} + E_{10} - E_{11} \in \mathbb{R}. \quad (18.2)$$

Note the sign on the last term, and note that in **CHSHStatisticalBridge.v** the same statistic is assembled with the $(1, 1)$ arguments reversed, giving $E_{11}^* = (\text{diff}_{11} - \text{same}_{11})/N_{11}$, so that all four terms may be added. The two presentations agree.

The classical bound is $|S| \leq 2$ and the Tsirelson bound is $|S| \leq 2\sqrt{2}$. Both are proved here as *algebraic* facts with no physics axioms. Where the development states the Tsirelson bound in squared form it appears as $S^2 \leq 8$, which is the same bound: $\sqrt{8} = 2\sqrt{2}$.

A correlation tuple $(E_{00}, E_{01}, E_{10}, E_{11})$ is **slice-coherent** (equivalently, column-contractive; the predicate tests slice membership rather than truthfulness, and the name records that) if:

$$1 - E_{00}^2 - E_{10}^2 \geq 0 \quad (\text{condition 1}) \quad (18.3)$$

$$1 - E_{01}^2 - E_{11}^2 \geq 0 \quad (\text{condition 2}) \quad (18.4)$$

$$(1 - E_{00}^2 - E_{10}^2)(1 - E_{01}^2 - E_{11}^2) \geq (E_{00}E_{01} + E_{10}E_{11})^2 \quad (\text{condition 3}) \quad (18.5)$$

These are exactly the PSD conditions on the 5×5 zero-marginal NPA moment matrix.

Theorem 18.1 (Slice Coherence $\iff$ NPA Realizability (coq/kernel/quantum/QuantumPartitionPSD.v, corollary column_contractive_iff_npa_psd), (S)). *column_contractive_iff_npa_psd*:

$$\text{slice_coherent}(\mathbf{E}) \iff \text{npa_psd}(\mathbf{E})$$

where *slice_coherent* means the three column-contractivity conditions hold, and *npa_psd* means the 5×5 zero-marginal NPA moment matrix is positive semidefinite.

Forward direction (*zero_marginal_npa_column_contractive_implies_psd* in coq/kernel/nfi/MuLedgerQuantumBridge.v): *column-contractive* $\Rightarrow$ *NPA PSD*. Proved by constructing the Cholesky decomposition of the 5×5 matrix from the three conditions.

Reverse direction (*npa_psd_implies_column_contractive* in coq/kernel/quantum/QuantumPartitionPSD.v): *NPA PSD* $\Rightarrow$ *column-contractive*. Proved by specializing the PSD condition at the vector $(0, -(E_{00}y_0 + E_{01}y_1), -(E_{10}y_0 + E_{11}y_1), y_0, y_1)$, obtaining a non-negative quadratic in y_0, y_1 , then closing via *quadratic_nonneg_discriminant*.

This biconditional has no project-local axioms; its assumption report includes standard-library real-number and functional-extensionality assumptions. No physics assumptions. Pure polynomial arithmetic. The zero-marginal NPA framework is the orthogonal-observables slice of the quantum elliptope ($\rho_{AA} = \rho_{BB} = 0$), not the full elliptope.

Key consequences:

- *PR box* ($E_{00} = E_{01} = E_{10} = 1, E_{11} = -1$) fails condition 1 immediately ($1-1-1 = -1 < 0$). Direct arithmetic; no separate theorem needed.
- *state_column_contractive_implies_tsirelson* (coq/kernel/nfi/MuLedgerQuantumBridge.v) directly proves *slice-coherent* $\Rightarrow |S|^2 \leq 8$ (equivalently $|S| \leq 2\sqrt{2}$); its proof composes *state_column_contractive_implies_npa_gram* with *npa_psd_implies_tsirelson_bound*. The Cauchy-Schwarz inner step (*PSD* $\Rightarrow$ row bounds $\Rightarrow |S|^2 \leq 8$) is *tsirelson_from_row_bounds* in coq/kernel/quantum/TsirelsonGeneral.v. The contrapositive ($|S| > 2\sqrt{2} \Rightarrow$ not slice-coherent) follows immediately. The converse is false and is not claimed.
- *Turing-classical correlations* $(1, 0, 1, 0)$ satisfy $|S| = 2$ but fail condition 1: $1-1-1 = -1 < 0$. A Turing machine can output this correlation; the slice gate rejects it as slice-incoherent (the full-elliptope gate of the next section accepts it, the point is inside the elliptope, outside the zero-marginal slice). Direct arithmetic from *column_contractive_iff_npa_psd*.

Zero Admitted. Zero named hypotheses.

18.2 The Elliptope Completion and Its Gate

The biconditional above characterizes the zero-marginal slice. The full correlator quantum set is characterized in coq/kernel/quantum/ElliptopeCompletion.v as an existential completion: $(E_{00}, E_{01}, E_{10}, E_{11})$ is **elliptope-realizable** (*elliptope_realizable*) iff some assignment of the cross moments $x = \langle A_0 A_1 \rangle$, $y = \langle B_0 B_1 \rangle$ makes the completed 5×5 moment matrix PSD.

Theorem 18.2 (Elliptope completion, kernel-tier). *All of the following hold with Print Assumptions reporting only the stdlib classical-reals and functional-extensionality families:*

- *deterministic_strategy_elliptope*: every deterministic sign strategy is elliptope-realizable, witnessed by its own cross moments (rank-one Gram argument).
- *elliptope_finite_mixture / lhu_mixture_elliptope*: the set is closed under arbitrary finite sub-convex mixtures; every LHV correlator is inside.

- *elliptope_tsirelson*: every member satisfies $S^2 \leq 8$, via Cauchy–Schwarz inside the PSD form (*psd_cauchy_schwarz*); row-norm bounds provably fail on the full set.
- *pr_box_not_elliptope*: the PR box admits no completion.
- *beyond_classical_elliptope*: $S = 12/5 > 2$ is attained inside, so the classical inclusion is strict.

The decidable gate (`coq/kernel/quantum/ElliptopeGate.v`) is the $\mathbb{Z}$ -arithmetic check an opcode can branch on: a fraction-free Sylvester branch over supplied completion buckets (strict interior) and a rational LDL^T -certificate branch reaching singular and boundary completions, with soundness (`elliptope_check_full_sound`) into `elliptope_realizable`, computed acceptances of the $\mu = 0$ tightness witness, $(1, 0, 1, 0)$, and the on-Tsirelson-curve point $(3/5, 4/5, 4/5, -3/5)$ (`gate_accepts_pythagorean_boundary`), and the universal refusal `elliptope_full_gate_never_accepts_pr_box`. Scope: correlator level; the only points beyond any exact integer check are those whose every PSD completion is irrational; the runtime opcode binding is future engineering.

18.3 The Pointer-Observable Criterion

`coq/kernel/frontier/PointerObservable.v` formalizes the schema behind the metering conjecture. An **ecosystem** (`Ecosystem`) is a state type carrying n observers, each holding a fragment map over states; an event E **redundantly proliferates** (`redundantly_proliferating`) iff every observer’s fragment records E ; and E is the **unique pointer among** a designated list of rival events (`unique_pointer_among`) iff E proliferates and no listed rival does. Uniqueness is relative to the named rivals on purpose, absolute uniqueness is false for free. A toy replicated-ledger instance shows the definitions are satisfiable and discriminating: the certification event is the unique pointer among the toy’s two meterable events (`toy_cert_unique_pointer`), a sanity instance only.

`PointerObservableReductions.v` instantiates the schema at models of the five metering disciplines of the reductions tier: proof-of-stake finality, gas metering, TEE attestation, certificate transparency, and proof-carrying verification. In each model every observer returns the selected flag, and a named rival flag is omitted by construction; all five are one construction with the fields renamed. The metered event is proved the unique pointer among its named rivals in each model (`PoS_model_unique_pointer`, `Gas_model_unique_pointer`, `TEE_model_unique_pointer`, `CT_model_unique_pointer`, `PCC_model_unique_pointer`), and the conjunction is exported as `five_labeled_models_have_selected_pointer`, which closes under the global Coq context, no axioms at all, `stdlib` or otherwise.

Scope: these theorems verify that the five formalized ecosystems have the claimed pointer structure. No protocol semantics, deployment, or security property occurs in their statements. Whether the real disciplines have that observer structure is a modeling claim argued in prose; if it holds, it is evidence for the pointer-observable conjecture, not a proof of it. The forced-event question stays open, and the file header says so.

18.3.1 The adversarial search

Five confirmations settle nothing on their own. A criterion loose enough to accept anything accepts five things just as easily, and what makes a universality class convincing is never the confirmations. It is the reported absence of exceptions after somebody went looking. `coq/kernel/frontier/PointerObservableCounterexamples.v` is the looking, and it is aimed at the corner where the criterion is most likely to break rather than at systems resembling the five.

The claim put at risk is the strong reading:

(PO-STRONG) Every system that must resist forgery of a commitment has a commitment event that redundantly proliferates.

A counterexample must satisfy three conditions fixed in advance: (C1) it must resist forgery of a commitment, in the sense that its designers treat an unearned commitment as the primary failure mode; (C2) it must be deployed and independently evolved, owing nothing to this development; and (C3) its commitment event must fail **redundantly_proliferating**.

Each candidate has two halves. (C1) and (C2) are facts about a real design and are argued in prose. The Coq half fixes an observer map and computes (C3) for it. The file contains no definition of forgery resistance, EUF-CMA security, deniability, memory safety, metering, or deployment, so the Coq results alone neither refute nor confirm a security claim.

(PO-STRONG) is false, on that two-part argument. Three candidates refute it; for each, the non-proliferating observer map is checked in Coq and its faithfulness to the design is argued in prose:

- **Deniable (designated-verifier) authentication** (`deniable_authentication_model_not_proliferating`). The scheme is engineered so a third party cannot be convinced even given the whole transcript, because the designated verifier could have produced it. Non-transferability is the product requirement.
- **Symmetric-key authentication** (`mac_model_not_proliferating`). Existential unforgeability is the defining security notion of a MAC, and verification requires the shared secret, so no record reaches anyone else.
- **Object capabilities** (`capability_model_not_proliferating`). Unforgeability here is unconditional, enforced by memory safety rather than by a cost or a hardness assumption. Nothing is paid, nothing is logged, and no observer outside the holder can tell.

A public-log control (`public_log_model_proliferating`) checks that the search can return *confirms* and is not an instrument that only reports refutations; its rival effort event still fails to proliferate (`public_log_effort_not_proliferating`), so the control discriminates rather than accepting everything.

The boundary these trace is sharp, and it is not the one (PO-STRONG) guessed at. A MAC and a digital signature both resist forgery; only the signature proliferates. Deniable authentication and certificate transparency both resist forgery; deniability actively prevents third-party conviction while transparency actively requires it. Object capabilities and proof-carrying code both make unearned authority impossible; the capability confines it to the holder while the proof travels to every consumer. In each pair the proliferating member is the one whose commitment must convince parties who were not present. The criterion that survives is therefore:

(PO-PUBLIC) A system whose commitment must convince third parties who did not witness its creation will proliferate records of the commitment event.

The signature observer model makes the commitment visible in each observer fragment and proves proliferation for that model. Its correspondence to deployed signatures remains a modeling judgment. Public availability of a verifier does not imply that every observer retains a record.

The verifier constructions give three sufficient evidence interfaces under their respective premises: a full-state transcript, an exact commitment-bit contract, and a reported response. Neither these examples nor the factorization obstruction proves that only one route can involve metering.

Three statements travel together in loose prose and come apart under these candidates:

(M1) Forgery resistance forces metering. False, on the prose argument. Signatures take the hardness route; object capabilities achieve unforgeability unconditionally through memory safety, with no cost, no record, and no assumption.

- (M2) **Forgery resistance forces record proliferation. False**, on the same argument. Deniable authentication, symmetric MACs, and capabilities all resist forgery while proliferating nothing, and in the first case that is the design goal.
- (M3) **Public commitments force record proliferation. Proposed.** The chosen observer models satisfy it. Actual record proliferation needs independent evidence; public verifiability alone is insufficient.

What remains of the metering claim, stated so a stranger can attack it: when a claim is μ -sensitive, meaning it is not decidable from the classical projection, and the verifier can neither recheck it directly nor substitute a hardness assumption, the substrate route is the available construction, and that route prices the commitment event. That is `V_does_not_factor_through_classical`, under its supplied colliding transcripts, in different clothes. It is narrower than “forgery resistance,” and it is the version this development supports.

Chapter 19

Spacetime Analogies: a μ -Pseudometric, Reachability Cones, and Discrete Geometry

19.1 The μ -Metric

Definition 19.1 (μ -Distance (MuGeometry.v)). The Coq formalization defines `mu_distance` as the absolute difference of μ -totals:

$$d_\mu(S_1, S_2) = |\mu_{\text{total}}(S_2) - \mu_{\text{total}}(S_1)|$$

This is a one-dimensional projection, not a minimum over paths.

Theorem 19.1 (Pseudometric Properties (MuGeometry.v)). *d_μ satisfies non-negativity, reflexivity ($d(S, S) = 0$), symmetry, and triangle inequality, all inherited from absolute value on $\mathbb{Z}$. Note: d_μ is a **pseudometric**, not a metric: $d(S_1, S_2) = 0$ does not imply $S_1 = S_2$ (many distinct states share the same μ -total). The “geometry” is a 1D projection onto the μ counter.*

19.2 Causal Cones

Definition 19.2 (μ -Reachability Cone (SpacetimeEmergence.v)). $C^+(S) = \{S' \mid \exists \tau. S \xrightarrow{\tau} S' \wedge \text{cost}(\tau) \leq \text{Budget}\}$.

This is a structural analogy to causal cones: the set of reachable states given a finite μ -budget plays a role analogous to a light cone. The analogy should not be over-read: there is no Lorentz invariance, no metric signature, and no boost transformations in the Coq formalization.

19.3 No-Signaling (Data Isolation)

Theorem 19.2 (Observational Locality (SpacetimeEmergence.v)). *If module M_B is not in the target set of any instruction in a trace, then M_B ’s observable region is unchanged after execution (`exec_trace_no_signaling_outside_cone`).*

Remark 19.1. This is a **data isolation** property: operations targeting module A in the partition graph do not affect the lookup of module B . The proof is structural (list operations on disjoint keys), non-trivial mainly for `instr_psplitt/instr_pmerge` which restructure the graph. The label “no-signaling” is an analogy. There is no spatial metric or speed limit between modules.

19.4 Discrete Geometry: A Restricted Angle-Defect Identity

The result is a conditional combinatorial identity with an assigned equilateral angle model. No physical interpretation is needed, but its graph, incidence, and boundary-count hypotheses are essential.

The formal statement is in `coq/kernel/curvature/DiscreteGaussBonnet.v`.

Theorem 19.3 (Restricted Angle-Defect Identity, (S)). *For a partition graph g satisfying `well_formed_triangulated`, assign $\pi/3$ to every triangle corner. The predicate includes $3F = 2I + B$, $\sum_v \deg(v) = 3F$, and $B = 3\chi$, where $\chi = V - E + F$; the edge definitions give $E = I + B$. With $\deg(v)$ counting incident triangles, define*

$$D(g) = \sum_v (2\pi - \deg(v)\pi/3).$$

Then $D(g) = 5\pi\chi(g)$. The definition uses 2π at every vertex, including boundary vertices.

Remark 19.2 (Hypotheses and relation to Gauss–Bonnet). The incidence equations yield $D = 2\pi V - \pi F = 2\pi\chi + \pi B$. The extra hypothesis $B = 3\chi$ supplies the coefficient 5π . It is not a general property of planar triangulations with boundary: a quadrilateral disk divided into two triangles has $B = 4$ and $\chi = 1$ and is excluded. The predicate also does not itself provide a planar embedding or a proof that the graph is a topological disk.

For a triangulated surface with boundary cycles, the usual combined interior-curvature and boundary-turning sum uses 2π at interior vertices and π at boundary vertices. Since boundary vertices and edges then have equal counts, this correction subtracts πB and gives $2\pi\chi$. Establishing the appropriate surface conditions and boundary-vertex definition is required for that statement. It is not the theorem formalized by the current file.

Thus the existing proof establishes a valid identity for its defined quantity under its stated restrictions. It neither derives those restrictions from computation nor establishes a physical curvature law or an angle scale from μ -cost.

19.4.1 Cross-link in MuGravity Vocabulary

Theorem 19.4 (Graph-Laplacian Sum-Zero Identity (`coq/kernel/frontier/F3_MuLaplacianSum.v`), (S)). *For every $VMState$ s :*

$$\sum_{m \in \text{pg_modules}(\text{vm_graph } s)} \text{mu_laplacian}(s, m) = 0.$$

Proof by edge-pair antisymmetry. `nat_list_disjoint` is symmetric, hence `modules_adjacent_by_region` is symmetric, hence `edge_weight` is symmetric; `mu_gradient` is antisymmetric by definition; the double sum cancels under the Fubini swap. No well-formedness, no calibration. Standalone graph-theoretic fact.

Theorem 19.5 (F3 Narrow Cross-link (`coq/kernel/frontier/F3_PartitionTopologyCrossLink.v`), (S)). *For every $VMState$ s at universal local calibration ($\forall m, \text{calibration_residual}(s, m) = 0$):*

$$\sum_{m \in \text{pg_modules}(\text{vm_graph } s)} \text{geometric_angle_defect}(s, m) = 0.$$

The proof composes two kernel facts: `calibration_residual_zero_iff` (geometry $\leftrightarrow \mu$ -ledger at every module) and `total_mu_laplacian_zero` (Theorem 19.4). The scalar examples `unit_residual_is_nonzero` and `pi_times_unit_is_nonzero` state that 1 and π are nonzero. They do not supply VM counter-instances establishing independence of the premises.

Remark 19.3 (Why this is not the topology version). The discrete angle-defect identity does not imply $\sum_m \mu_{\text{laplacian}} = 5 \cdot \chi$. That inference would require identifying two distinct angle-defect quantities: `DiscreteGaussBonnet.angle_defect` (degree $\times \pi/3$, equilateral) and `MuGravity.geometric_angle_defect` (distance-weighted, $\pi \cdot d_{bc}/(1 + d_{ab} + d_{ac} + d_{bc})$). They are not equal in general: `strong_bridge_counterexample` (in the same file) formalises a degree-6 vertex at uniform inter-module distance $d \geq 1$ where the Gauss-Bonnet defect is 0 and the MuGravity defect is $2\pi/(1+3d) \neq 0$. The cross-link makes no claim about the topology invariant. Discrete Gauss-Bonnet remains a kernel theorem; it is simply not load-bearing for this cross-link.

Non-vacuity is open. Symbolic search across small actually-triangulated graphs (K_4 , $K_4 - e$, octahedron, prism, K_5 , B_3 , K_6 , K_7) up to $N = 7$ returns no integer-mass calibrated solution; structural analysis identifies three failure modes (insufficient triangles, excess triangles, asymptotic-only target reachability). `module_triangles` was tightened in `MuGravity.v` to require pairwise-adjacent neighbors (an actual graph triangle), eliminating tree-shaped pseudo-witnesses.

19.5 Four-Dimensional Discrete Riemannian Geometry (Reference Only, Not a Main Claim)

Remark 19.4 (This section is not about spacetime). The metric defined below is diagonal with entries $2 \cdot |\text{module_structural_mass}|$, which are non-negative, so the signature is $(+, +, +, +)$: the form is *Riemannian*, not Lorentzian. There is no timelike direction, no light cone, no causal structure, and no proper time. Calling a positive-definite 4-dimensional metric “spacetime” misdescribes the object even under a reference-only label.

Read this section as: discrete Riemannian geometry on a 4-dimensional simplicial complex built from partition-graph data. Every general-relativistic word that appears below (Christoffel, Riemann, Einstein, stress-energy) is being used for the corresponding *Riemannian* construction. The vacuum result remains a $0 = 0$ consistency check, as the scope warning says.

Remark 19.5 (Scope Warning). The following subsections define discrete differential geometry objects (simplicial complex, metric tensor, Christoffel symbols, Riemann tensor, Einstein tensor) over partition graphs and prove results about the vacuum case $G_{\mu\nu} = 0 = T_{\mu\nu}$. This is a $0 = 0$ result: both sides are zero in the vacuum because mass is zero everywhere. It is **not** a derivation of Einstein’s field equations from computation. The gravitational constant $G := 1/(8\pi)$ is a unit choice (a definition), not a derived quantity. No physical claim is being made. The geometry section’s main result is the Discrete Gauss-Bonnet theorem (Section 19.4). Everything in this section is marked reference-only and does not appear in the primary claim ledger.

The previous sections treated spacetime as a 1D projection via the μ -metric. The next subsections extend to full 4D geometry as a reference exploration. None of the results here are part of the core formal claims of the project.

19.5.1 4D Simplicial Complex

Definition 19.3 (4D Simplicial Complex (`FourDSimplicialComplex.v`)). A **4D simplicial complex** $\mathcal{K}$ is a discrete approximation to 4D spacetime consisting of:

- **Vertices** V : Computational modules (`ModuleID`)
- **Edges** $E \subseteq V \times V$: Connections between modules
- **Simplices**: 2-simplices (triangles), 3-simplices (tetrahedra), 4-simplices (4D cells)

The complex satisfies:

- **Consistency:** Every face of a simplex is also in $\mathcal{K}$
- **Well-formedness:** Dimension 4 with locally Euclidean structure

19.5.2 Metric Tensor from μ -Costs

Definition 19.4 (Computational Metric (MetricFromMuCosts.v, RiemannTensor4D.v)). The metric tensor at vertex v with respect to vertices w_1, w_2 is:

$$g_{\mu\nu}(v; w_1, w_2) = \begin{cases} \ell(v, w_1, w_2) & \text{if } \mu = \nu \\ 0 & \text{if } \mu \neq \nu \end{cases} \quad (19.1)$$

where the edge length function is:

$$\ell(s, v_1, v_2) = 2 \cdot \text{mass}(v_1) \quad \text{if } v_1 = v_2 \quad (19.2)$$

and mass is derived from structural complexity: $\text{mass}(v) = |\text{module_structural_mass}(s, v)|$.

Remark 19.6 (The metric is specified only on the diagonal). ℓ is given only for $v_1 = v_2$. No value is stated for $v_1 \neq v_2$, so as written g is a partial specification. This matters downstream rather than cosmetically: the Christoffel symbols require $\partial_\mu g_{\nu\sigma}$, and the discrete derivative *differences across neighbouring vertices*. It evaluates g exactly where the definition above is silent. The Coq development supplies a total function, so the mechanised objects are well-defined; this document simply does not state which extension it uses. Treat the equations below as schematic until the off-diagonal case is written down here. Since the whole section is reference-only and the vacuum result is a $0 = 0$ check, nothing in the claim ledger depends on the resolution.

Also note g is positive semi-definite, so it degenerates wherever $\text{mass}(v) = 0$, including the entire vacuum configuration, which is the case the theorems below actually evaluate. A degenerate metric has no inverse, so “raising an index” is not available there.

Remark 19.7. This metric is **position-dependent** for non-uniform mass distributions:

- **Uniform mass m :** $g_{\mu\mu}(v; w, w) = 2m$ (constant) $\implies$ flat spacetime
- **Non-uniform mass:** $g_{\mu\mu}(v; w, w)$ varies with $w \implies$ curved spacetime

In this model, mass distributions create position-dependent geometry: uniform mass gives flat spacetime, non-uniform mass gives curved spacetime.

19.5.3 Discrete Differential Geometry

Definition 19.5 (Discrete Derivative (RiemannTensor4D.v)). For a scalar field $f : V \rightarrow \mathbb{R}$ and direction μ :

$$\partial_\mu f(v) = \frac{1}{|N_\mu(v)|} \sum_{w \in N_\mu(v)} (f(w) - f(v)) \quad (19.3)$$

where $N_\mu(v)$ is the set of neighbors of v in the μ -direction.

Definition 19.6 (Christoffel Symbols (RiemannTensor4D.v)). The connection coefficients (Christoffel symbols of the second kind) are:

$$\Gamma_{\mu\nu}^\rho(v) = \frac{1}{2} \sum_\sigma g^{\rho\sigma}(v) (\partial_\mu g_{\nu\sigma}(v) + \partial_\nu g_{\mu\sigma}(v) - \partial_\sigma g_{\mu\nu}(v)) \quad (19.4)$$

where $g^{\rho\sigma}$ is the inverse metric (diagonal in this construction).

Lemma 19.6 (Flat Spacetime Has Zero Connection (EinsteinEquations4D.v), (S)). *For uniform mass distribution:*

$$\forall w, \text{mass}(w) = m \implies \Gamma_{\mu\nu}^\rho(v) = 0 \quad (19.5)$$

Proof. Uniform mass $\implies$ position-independent metric $\implies \partial_\mu g_{\nu\sigma} = 0 \implies \Gamma_{\mu\nu}^\rho = 0$. $\square$

19.5.4 Curvature Tensors

Definition 19.7 (Riemann Curvature Tensor (RiemannTensor4D.v)). The Riemann curvature tensor measures the failure of parallel transport:

$$R_{\sigma\mu\nu}^\rho(v) = \partial_\mu \Gamma_{\nu\sigma}^\rho(v) - \partial_\nu \Gamma_{\mu\sigma}^\rho(v) + \sum_\lambda \left(\Gamma_{\mu\lambda}^\rho(v) \Gamma_{\nu\sigma}^\lambda(v) - \Gamma_{\nu\lambda}^\rho(v) \Gamma_{\mu\sigma}^\lambda(v) \right) \quad (19.6)$$

Definition 19.8 (Ricci Tensor (RiemannTensor4D.v)). Contraction of the Riemann tensor:

$$R_{\mu\nu}(v) = \sum_\rho R_{\mu\rho\nu}^\rho(v) \quad (19.7)$$

Definition 19.9 (Ricci Scalar (RiemannTensor4D.v)). Complete contraction:

$$R(v) = \sum_{\mu,\nu} g^{\mu\nu}(v) R_{\mu\nu}(v) \quad (19.8)$$

Definition 19.10 (Einstein Tensor (RiemannTensor4D.v)). The Einstein tensor encodes space-time curvature:

$$G_{\mu\nu}(v) = R_{\mu\nu}(v) - \frac{1}{2} g_{\mu\nu}(v) R(v) \quad (19.9)$$

Lemma 19.7 (Flat Spacetime Has Zero Curvature (EinsteinEquations4D.v), (S)). *For uniform mass:*

$$\forall w, \text{mass}(w) = m \implies R_{\sigma\mu\nu}^\rho(v) = 0 \implies G_{\mu\nu}(v) = 0 \quad (19.10)$$

19.5.5 Stress-Energy Tensor

Remark 19.8 (The components below are named by analogy, not by physics). The three definitions that follow assign the letters T_{00} , T_{0i} , T_{ij} to expressions in `module_structural_mass`. The names are borrowed from the relativistic stress-energy tensor; the objects are not those quantities, and three specific mismatches should be stated plainly rather than left for a reader to discover:

1. **T_{00} is not an energy density.** It is set equal to $\text{mass}(v)$, a dimensionless integer attached to a vertex. An energy density is energy per unit volume; there is no volume element anywhere in this construction, and no conversion constant. Compare Section 21.7: a dimensionless integer ledger cannot produce a dimensional constant, which is exactly the obstruction here.
2. **T_{0i} is not a momentum density.** Momentum density is ρv^i , mass density times velocity. The definition below is $\partial_i T_{00}$, the spatial *gradient* of the energy-density slot. That is a different object with different index structure and different dimensions; a gradient of a density is not a flux. There is no velocity field in this model for it to be built from.
3. **$T_{ij} = \delta_{ij} \cdot \text{mass}$ silently fixes an equation of state.** An isotropic spatial block with pressure equal to energy density is $p = \rho$: a *stiff fluid*, the maximally rigid equation of state consistent with causality in the relativistic setting. Nothing in the model motivates that choice; it is a definitional convenience, and it should not be read as a derived result about the matter content.

None of this affects the vacuum theorem, where every component is 0 because $\text{mass} \equiv 0$. That is also why these mismatches have never mattered computationally. They matter for how the section is read.

Definition 19.11 (T_{00} slot: vertex mass (EinsteinEquations4D.v)). The 00 component is set to the computational mass. Per Remark 19.8 this is not an energy density:

$$T_{00}(v) = \text{mass}(v) \quad (19.11)$$

Definition 19.12 (T_{0i} slot: spatial gradient of T_{00} (EinsteinEquations4D.v)). The mixed components are set to the spatial derivative of the 00 slot. Per Remark 19.8 this is a gradient, *not* a momentum density:

$$T_{0i}(v) = T_{i0}(v) = \partial_i T_{00}(v) \quad (19.12)$$

Definition 19.13 (T_{ij} slot: isotropic, $p = \rho$ (EinsteinEquations4D.v)). The spatial block is isotropic with diagonal entries equal to the 00 slot. Per Remark 19.8, setting the pressure equal to the energy-density slot is the stiff-fluid equation of state $p = \rho$; it is imposed here by definition, *not* derived:

$$T_{ij}(v) = \begin{cases} \text{mass}(v) & \text{if } i = j \\ 0 & \text{if } i \neq j \end{cases} \quad (19.13)$$

Definition 19.14 (Stress-Energy Tensor (EinsteinEquations4D.v)). Combining all components:

$$T_{\mu\nu}(s, \mathcal{K}, v) = \begin{cases} T_{00}(v) & \text{if } \mu = \nu = 0 \\ T_{0i}(v) & \text{if } \mu = 0, \nu = i \text{ or } \mu = i, \nu = 0 \\ T_{ij}(v) & \text{if } \mu = i, \nu = j \\ 0 & \text{otherwise} \end{cases} \quad (19.14)$$

19.5.6 Einstein's Field Equations

Definition 19.15 (Gravitational Constant (EinsteinEquations4D.v)). In computational units where the fundamental scale is set by μ -costs:

$$G = \frac{1}{8\pi} \quad (19.15)$$

This makes $8\pi G = 1$, normalizing the coupling in Einstein's equations.

Lemma 19.8 (Curvature from μ -Gradients (EinsteinEquations4D.v), (S)). *For uniform mass distribution:*

$$\forall w, \text{mass}(w) = m \implies G_{\mu\nu}(v) = 0 \quad (19.16)$$

Proof. The proof chain is:

1. Uniform mass $\implies$ position-independent metric
2. Position-independent metric $\implies$ zero discrete derivatives $\partial_\mu g_{\nu\sigma} = 0$
3. Zero metric derivatives $\implies$ zero Christoffel symbols $\Gamma_{\mu\nu}^\rho = 0$
4. Zero Christoffel $\implies$ zero Riemann tensor $R_{\sigma\mu\nu}^\rho = 0$
5. Zero Riemann $\implies$ zero Ricci tensor and scalar
6. Zero Ricci $\implies$ zero Einstein tensor $G_{\mu\nu} = 0$

All steps are proven constructively in EinsteinEquations4D.v. □

Lemma 19.9 (Vacuum Implies Zero Stress-Energy (EinsteinEquations4D.v), (S)). *For vacuum states (zero mass everywhere):*

$$\forall w, \text{mass}(w) = 0 \implies T_{\mu\nu}(v) = 0 \quad (19.17)$$

Proof. Direct computation:

- $T_{00} = \text{mass} = 0$
- $T_{0i} = \partial_i(\text{mass}) = \partial_i(0) = 0$
- $T_{ij} = \delta_{ij} \cdot \text{mass} = \delta_{ij} \cdot 0 = 0$

□

Theorem 19.10 (Vacuum Case: Both Sides Are Zero (`coq/kernel/curvature/EinsteinEquations4D.v`), (**R**)). *For any VM state s , 4D simplicial complex $\mathcal{K}$, space-time indices μ, ν , and vertex v :*

$$\forall w, \text{mass}(w) = 0 \implies G_{\mu\nu}(s, \mathcal{K}, v) = 8\pi G T_{\mu\nu}(s, \mathcal{K}, v) \quad (19.18)$$

Honest status: *this is 0 = 0. Both sides equal zero in vacuum because all masses vanish. The gravitational constant $G := 1/(8\pi)$ is a definition (unit choice), not a derived quantity. This is **not** a derivation of Einstein’s field equations. It is a consistency check that the discrete geometric objects reduce correctly when mass is everywhere zero. The non-vacuum case is not proven. This theorem does not appear in the primary claim ledger of the project.*

Proof. Step 1: Expand the right-hand side using $G = 1/(8\pi)$:

$$8\pi G T_{\mu\nu} = 8\pi \cdot \frac{1}{8\pi} \cdot T_{\mu\nu} = T_{\mu\nu} \quad (19.19)$$

Step 2: Apply Lemma 19.8 (curvature from μ -gradients):

Vacuum (mass = 0 everywhere) is a special case of uniform mass distribution, so:

$$G_{\mu\nu}(v) = 0 \quad (19.20)$$

Step 3: Apply Lemma 19.9 (stress-energy conservation):

For vacuum:

$$T_{\mu\nu}(v) = 0 \quad (19.21)$$

Conclusion: Both sides equal zero:

$$0 = 0 \quad \checkmark \quad (19.22)$$

The complete proof is formalized in `coq/kernel/curvature/EinsteinEquations4D.v` with:

- No project-local axioms
- Zero admitted lemmas
- Zero classical logic imports
- Verified by automated proof auditing (Inquisitor: 0 HIGH findings)

□

Remark 19.9 (Vacuum vs. Non-Vacuum). Theorem 19.10 requires the vacuum hypothesis (mass = 0 everywhere). For non-vacuum states with arbitrary mass distributions, the equation $G_{\mu\nu} = 8\pi G T_{\mu\nu}$ only holds when the distribution satisfies discrete Bianchi identities (conservation laws). This is analogous to classical general relativity, where Einstein’s equations are *field equations* that constrain which configurations are physically possible, not universal identities satisfied by all states.

The vacuum construction is an internal consistency result. It does not establish that physical spacetime curvature emerges from computation. A non-vacuum physical derivation requires additional geometric and physical premises with independently justified applicability.

Remark 19.10 (Epistemological Status: **(R)** Consistency Relation). This is a **(R)** result: the vacuum theorem reduces to $0 = 0$ (both sides vanish when all masses are zero; the gravitational constant is a definition). It establishes that the discrete geometric objects are internally consistent in the zero-mass case. It does *not* establish that spacetime geometry emerges from computation in any physically meaningful sense; that would require the A_QM named hypothesis and is an open empirical question. The non-vacuum case (the full Einstein field equations for arbitrary mass distributions) is not proved and is not a claim of this project.

Remark 19.11. The discrete geometric objects (metric, curvature, Einstein tensor) are defined from computational primitives. In the vacuum case (all masses zero) they all reduce to zero. This is a consistency check, not a derivation of spacetime emergence. The claim that spacetime geometry is an emergent property of computation would require more: a non-trivial non-vacuum case, a connection to physical observations, and the named A_QM bridge. None of those are present here.

19.5.7 Empirical Validation

The theoretical results are validated numerically:

Theorem 19.11 (Numerical Verification (tests/test_*), Empirical).

1. **Flat spacetime** (uniform mass $m = 1$):

- $\max |\Gamma_{\mu\nu}^\rho| < 10^{-10}$ ✓
- $\max |R_{\sigma\mu\nu}^\rho| < 10^{-10}$ ✓
- $\max |G_{\mu\nu}| < 10^{-10}$ ✓

2. **Curved spacetime** (non-uniform mass $m \in [0.5, 1.5]$):

- $\max |\Gamma_{\mu\nu}^\rho| \approx 3.0$ ✓ (curvature detected)
- $\max |R_{\sigma\mu\nu}^\rho| > 0$ ✓

3. **Vacuum Einstein equations** (mass = 0):

- $|G_{\mu\nu}| < 10^{-10}$ ✓
- $|T_{\mu\nu}| < 10^{-10}$ ✓
- $|G_{\mu\nu} - 8\pi GT_{\mu\nu}| < 10^{-10}$ ✓

These tests confirm that the discrete differential geometry correctly implements the theoretical predictions.

19.5.8 Affine Metric-Scaled Operator: Non-Uniform Boundary Results

The first-neighbor discrete operator refutes the non-uniform diagonal extension of the EFE. A signed affine metric-scaled redesign closes the gap at vertex 1 and fully characterizes the outer vertices.

Theorem 19.12 (Affine Off-Diagonal Ricci Zero (kernel/AffineEFEClosure.v), **(S)**). *affine_off_diagonal_ricci_zero: Under the affine metric-scaled symmetric operator, with the non-uniform isotropic boundary 4-simplex metric, off-diagonal Ricci vanishes at vertex 1: $R_{\mu\nu}(v_1) = 0$ for $\mu \neq \nu$. Zero Admitted. Zero Section Variables.*

Theorem 19.13 (Full Tensor EFE at Vertex 1 (kernel/AffineEFEClosure.v), (S)). *affine_full_tensor_efe*: Under *module_structural_mass s 1 = 1* (unit structural mass at vertex 1), the full tensor Einstein field equation holds:

$$G_{\mu\nu}(v_1) = 3 \cdot T_{\mu\nu}(v_1)$$

for all $\mu, \nu \in \{0, 1, 2, 3\}$. Zero Admitted.

Theorem 19.14 (Outer Vertex Characterization (kernel/AffineEFEClosure.v), (S)). At the outer vertices $v \in \{0, 2, 3, 4\}$ of the boundary 4-simplex, the affine operator produces:

- *affine_ricci_offdiag_outer*: Off-diagonal Ricci = $3/32$ (non-zero).
- *affine_ricci_diag_outer*: Diagonal Ricci = $-45/32$.
- *affine_ricci_scalar_outer*: Ricci scalar = $-45/8$.
- *affine_einstein_outer*: Einstein tensor = $45/32$ (diagonal) and $3/32$ (off-diagonal).
- *affine_efc_fails_outer_offdiag*: The diagonal EFE form $G = 3T$ fails at outer vertices because G has non-zero off-diagonal components.

The affine factor $8 - 5g_{00}$ is specifically tuned to the center vertex geometry; at outer vertices it produces a non-trivial off-diagonal Einstein tensor. Zero Admitted.

Remark 19.12 (What Remains Open). The affine operator closes vertex 1 and characterizes the outer vertices. The remaining open question is whether the affine factor $8 - 5g_{00}$ is the correct physical discretization. That requires connecting the discrete operator to a physical substrate, not a Coq question, an empirical one.

Chapter 20

Information Causality

Theorem 20.1 (IC- μ Equivalence (InformationCausality.v), (R)). *The file defines two record types (`ICScenario` and `MuScenario`) with opaque `Prop` fields, then proves they are equivalent given a hypothesis (`ic_mu_equivalent`) that literally states the equivalence. The proof is `tauto`. No mutual information, probability distributions, or CHSH values appear anywhere in the formalization.*

Remark 20.1. The intended statement, that Bob’s total information about Alice’s data is bounded by the channel’s μ -capacity, is outside the formalized result. The Coq file packages the proposed equivalence as an explicit hypothesis and derives the stated consequence from that hypothesis. It defines neither entropy nor mutual information and proves no bound from the VM semantics; those physical quantities and their connection to μ are outside the present formal scope.

Chapter 21

Thermodynamic Bridge: Conditional Links and Their Limits

21.1 Landauer’s Principle

Theorem 21.1 (Landauer-Shaped Erasure Arithmetic (LandauerDerived.v), (C)).

1. *erasure_irreversible*: Erasing ≥ 1 bit implies fan-in = $2^k > 1$ (arithmetic: $2^k > 1$ for $k \geq 1$).
2. *erasure_decreases_entropy*: Erasure decreases system entropy ($\Delta S < 0$), this is just $\text{output} < \text{input} \Rightarrow \text{output} - \text{input} < 0$.
3. *landauer_information_bound*: For any physical erasure, environment entropy increase $\geq$ bits erased. Here a physical erasure is a *PhysicalErasure* record, which carries this bound as a field (see the caveat).
4. *erasure_additive*: Sequential erasures compose additively.

Caveat on item 3: The *PhysicalErasure* record has *second_law_satisfied* as a **field**, not a derived property. The theorem *landauer_information_bound* simply extracts this field, making the bound an input assumption rather than a derivation. The physical energy bound $Q \geq k_B T \ln 2 \cdot n$ is **explicitly not proven** (the file itself notes the conversion factor is not formalized). Also, the *mu* in this file is a standalone synonym for *bits_erased*, it has no formal connection to *vm_mu* in *VMState.v*.

21.2 A2 from Landauer (Factored Form): a Conditional Whose Premises Fail on the Full ISA

Theorem 21.2 (F1 Strong Form: A2 from Two Landauer-Style Premises (coq/kernel/frontier/F1_StrongForm.v), (C)). Given an abstract dissipation function $\phi : \text{vm_instruction} \rightarrow \mathbb{N}$ (interpreted as dissipation per step in Landauer quanta), if

1. **Landauer’s principle** holds in dissipation vocabulary, every class-collapsing step on any bool macro-property has $\phi(i) \geq 1$, and
2. the named bridge *cost_dissipation_calibrated* holds, $\text{instruction_cost}(i) \geq \phi(i)$ for every i ,

then on every cert-flip step $\text{instruction_cost}(i) \geq 1$. The Landauer-style premise mentions only class-collapse and dissipation; *instruction_cost* appears only in the calibration. This is a valid conditional implication, but its two premises are incompatible on the current full VM instruction set, as the following remark explains.

Remark 21.1 (Applicability failure of the full-ISA physical premises). `F1_physical_premises_incompatible` proves that no dissipation function satisfies both hypotheses. The instruction `JUMP(1,0)` has zero cost and sends every state’s program counter to 1. For the Boolean macro-property “PC equals 1,” it meets `step_collapses_bool_classes`: a state with PC 0 enters the true class, and states already in that class stay there. Hypothesis (1) therefore requires dissipation at least 1, while hypothesis (2) bounds it by 0.

Thus this theorem does not supply an applicable physical derivation of A2 for the current VM. A2 follows directly from its instruction semantics, without either physical premise. A physical account must justify a domain of operations and macro-properties, reconcile its dissipation model with the cost schedule, and exhibit jointly satisfiable premises. Merely losing a distinction under a selected macro-observation does not itself establish physical information erasure.

Remark 21.2 (Integrality of ϕ is a physical assumption, not a typing choice). ϕ is declared with codomain $\mathbb{N}$, and the Landauer premise is stated as $\phi(i) \geq 1$. The integrality is doing real work: it is what makes the floor come out as “at least one” rather than “at least $k_B T \ln 2$ ”. But Landauer’s principle gives a *real-valued* lower bound on dissipated energy per bit erased; nothing in thermodynamics says dissipation arrives in integer multiples of a quantum. Choosing $\mathbb{N}$ as the codomain therefore imports a quantisation assumption through the type signature, where it is easy to miss. It should be read as a named modelling assumption on the same footing as `cost_dissipation_calibrated`, not as a neutral formalisation detail.

Theorem 21.3 (Calibration consistency and retained history, (S)). *For eligible operations $E(i)$ and a schedule $c(i)$, a function d with $E(i) \Rightarrow d(i) \geq 1$ and $d(i) \leq c(i)$ exists exactly when all eligible operations have positive scheduled cost. The witness in the converse is $d = c$. A Boolean identity/reset model realizes the conditions with charges zero and one respectively.*

For any deterministic step function, the extension

$$(s, h) \mapsto (\text{step}(s, i), s :: h)$$

is injective for fixed i , has a left inverse on its image, and projects to the original transition. In particular, the zero-cost VM jump has an injective history lift. These results establish consistency of an abstract restricted model and failure of macro-collapse as a general erasure test. They do not identify the cost units with heat.

21.3 Permanent Certification Forces a Merge

The history lift escapes the merge by keeping unbounded history. On a finite state space with a certificate that no step revokes, that escape is closed for the certifying step.

Theorem 21.4 (Permanent flip is not injective (`coq/kernel/nfi/PermanentCertification.v`), (S)). *Let S have a duplicate-free enumeration, let $\text{step} : S \rightarrow I \rightarrow S$ and $\text{cert} : S \rightarrow \text{bool}$ be arbitrary, and suppose $\text{cert}(s) = \text{true} \Rightarrow \text{cert}(\text{step}(s, i)) = \text{true}$ for all s, i . If $\text{cert}(s) = \text{false}$ and $\text{cert}(\text{step}(s, i)) = \text{true}$, then $\text{step}(\cdot, i)$ is not injective (`permanent_flip_is_not_injective`). If k distinct uncertified states are switched on by one instruction, that instruction’s step has at least k fewer distinct images than inputs on the certified states together with those k (`permanent_flips_collapse_at_least`).*

Theorem 21.5 (A2 from a merging price (`PermanentCertification.v`), (C)). *Under the same premises, if every instruction whose step is not injective costs at least 1, then every certification flip costs at least 1 (`a2_from_merging_price_and_permanence`). The system is then a `CertificationSystem` with its A2 field discharged (`certification_system_from_merging_price`), and the trace floor follows (`permanent_certification_trace_floor`). Honest erasure accounting, whose flag reports every merging instruction, gives the same floor (`honest_erasure_accounting_implies_a2`); the zero-cost rival of `CommitmentVsErasure.v` is not honest in this sense (`commit_without_erasure_system_is_not_honest`).*

Remark 21.3 (Scope and necessity of each premise). The merging price is a named premise standing for Landauer’s principle; it is conditional in the sense of (C). The premises are jointly satisfiable (`priced_reset_satisfies_premises`). Each is needed: an unbounded history keeps every step injective while a permanent certificate switches on (`unbounded_history_escapes`); negation switches a revocable certificate on bijectively (`revocable_certificate_escapes`); an unpriced reset certifies at cost zero (`free_merge_escapes`). The 51-opcode VM is not an instance, by design: its ledger is unbounded, and its schedule prices the certifying merge while JUMP merges program counters at zero declared cost (`vm_prices_certifying_merge_leaves_others_free`, below). A finite fragment of it is an instance (`FiniteCertMachine.v`, below). All ten theorems are closed under the global context.

Remark 21.4 (Prior results used). The ingredients are standard: an unconditional set of a bit is Landauer’s example of a logically irreversible operation (1961); finite memory forcing erasure is Bennett’s observation on reversible computation; the core counting is the pigeonhole principle; the entropy support bound and the monotonicity of entropy under deterministic maps are Shannon’s; the dependence of Landauer’s bound on the input distribution is the generalized form of Maroney (2009) and Sagawa (2014). The contribution recorded here is the combination about permanent certificates, the merge-or-revoke characterization, and the mechanized statements. Shadow pricing (Section 21.4) is fiber constancy (`decoding_requires_fiber_constancy`) applied to prices.

Theorem 21.6 (Cost of a squeeze (`coq/kernel/nfi/PermanentRecordPricing.v`), (C)). *Assume the counting premise `compression_priced`: for every instruction i and duplicate-free list D , $|D| \leq 2^{\text{cost}(i)} \cdot |\text{nodup}(\text{step}(\cdot, i)[D])|$. On a finite state space with a permanent certificate and m certified states, if instruction i switches k distinct uncertified states on, then $m + k \leq 2^{\text{cost}(i)} m$ (`permanent_flips_compression_bound`) and $\lceil \log_2 \rceil(m + k) \leq \text{cost}(i) + \lceil \log_2 \rceil m$ (`permanent_flips_log_bound`). With $k = 1$ this yields A2 (`a2_from_compression_price_and_permanence`). A four-state stamp attains the bound with cost 2 for $k = 3$, $m = 1$ (`quad_stamp_needs_two`, `quad_stamp_cost_two_is_priced`).*

Theorem 21.7 (Forced price and the merge-or-revoke dichotomy (`PermanentRecordPricing.v`), (S)). *Call i forced to be priced when every cost assignment satisfying `merging_steps_priced` charges i at least 1. With decidable equality on instructions, i is forced to be priced iff $\text{step}(\cdot, i)$ is not injective (`forced_priced_iff_merges`). On a finite state space, if i switches a reading on at some state, then either $\text{step}(\cdot, i)$ is not injective or i switches the reading off at some certified state (`flip_merges_or_revokes`, proved constructively by a finite search). Hence an instruction that never revokes the reading and flips it somewhere is forced to be priced (`permanent_record_write_is_forced_priced`). The converse fails: a three-state instruction flips, revokes, and merges (`forced_price_without_permanent_record`). The trace-level floor follows through `universal_nfi_any_substrate` (`compression_priced_trace_floor`). All twelve theorems and corollaries in the file are closed under the global context.*

Theorem 21.8 (Entropy form (`coq/kernel/nfi/PermanentCertificationEntropy.v`), (S) for the bound, (C) for heat). *Let $H(p) = -\sum_s p(s) \log_2 p(s)$ over the enumeration, with $0 \log 0 = 0$, and let f_*p be the pushforward of p along $\text{step}(\cdot, i)$. Entropy is at most $\log_2$ of the size of any list covering the support (`entropy_le_log_support`, from $\ln y \leq y - 1$). On a finite state space with a permanent certificate, if i switches $k \geq 1$ distinct uncertified states on beside m certified ones and p is supported on those $m + k$ states, then $H(f_*p) \leq \log_2 m$ (`permanent_step_entropy_ceiling`); for p uniform on them, $H(p) - H(f_*p) \geq \log_2((m + k)/m)$ (`permanent_flip_uniform_entropy_drop`). An injective step leaves H unchanged (`step_entropy_invariant_if_injective`). A whole-unit cost with $\text{cost}(i) \geq H(p) - H(f_*p)$ for every distribution p satisfies A2 (`a2_from_entropy_price_and_permanence`), with the trace floor through `universal_nfi_any_substrate` (`entropy_priced_trace_floor`). Under the named premise `landauer_heat`*

($\text{heat} \geq k_B T \ln 2 \cdot \Delta H$), the uniform flip dissipates at least $k_B T \ln((m+k)/m)$ (*permanent_flip_heat_floor*), which is positive for $T > 0$ (*permanent_flip_heat_positive*). For any p and any deterministic f , $H(p) - H(f_*p) = \sum_{p(x) > 0} p(x)(\log_2(f_*p)(fx) - \log_2 p(x))$ (*entropy_drop_as_point_sum*), each term is nonnegative (*entropy_drop_nonneg*), and the sum is positive when $x \neq x'$, $fx = fx'$, $p(x) > 0$, $p(x') > 0$ (*entropy_drop_pos_of_support_merge*). Hence a permanent flip removes entropy for every p positive on the certified states and the flipping state (*permanent_flip_full_support_entropy_drop_positive*), with positive heat under the premise (*permanent_flip_full_support_heat_positive*). For a point mass the drop is 0 and the premise reduces to $\text{heat} \geq 0$ (*known_state_step_removes_no_entropy*, *known_state_flip_forces_no_heat*): the forced heat is a worst-case statement over the state distribution, as in the generalized Landauer bound. The file works over Coq's real numbers; its results depend on the standard-library constants *sig_forall_dec*, *sig_not_dec*, *functional_extensionality_dep*, and *classic*, and on nothing project-local.

Theorem 21.9 (A finite instance run by the VM (*coq/kernel/nfi/FiniteCertMachine.v*), (S)). Let $\text{FState} = \{L_0, L_1, L_2, L_3\} \times \text{bool}$ and $\text{FInstr} = \{\text{FCertify}, \text{FJump } a, \text{FNext}\}$ with $\text{FCertify}(p, c) = (p + 1, \text{true})$, $\text{FJump } a(p, c) = (a, c)$, $\text{FNext}(p, c) = (p + 1, c)$ (*slots modulo 4*), certificate π_2 , and costs 1, 2, 0. The eight states form a finite enumeration (*fin_finite*), the certificate is permanent (*fin_permanent*), *FNext* is injective and the other two are not, the costs satisfy *merging_steps_priced* (*fin_merging_priced*) and *compression_priced* (*fin_compression_priced*, via the fiber lemma *fiber_bound_compression*), and A2 follows (*fin_a2_from_merging_price*, *fin_a2_from_compression_price*). With window $w(s) = (\text{slot}(\text{vm_pc} \bmod 4), \text{vm_certified})$ and $\text{FCertify} \mapsto \text{CERTIFY } 0$, $\text{FJump } a \mapsto \text{JUMP } a \ 2$, $\text{FNext} \mapsto \text{CHECKPOINT "" } 0$: $w(\text{vm_apply}(s, \iota)) = \text{fst}(\text{step}(w(s), \iota))$ and μ rises by exactly $\text{fcost}(\iota)$ (*vm_runs_finite_machine*, *vm_pays_finite_price*), for whole runs as well (*vm_runs_finite_trace*); a run of the fragment from uncertified to certified raises μ by at least 1 (*vm_fragment_certification_paid*). The *JUMP* cost of 2 is a program's choice of μ_Δ . All twenty-three theorems and lemmas in the file are closed under the global context.

Theorem 21.10 (What the full VM prices (*FiniteCertMachine.v*), (S)). $\text{vm_apply}(\cdot, \text{CERTIFY } d)$ is not injective: states differing only in *vm_certified* have the same image (*vm_certify_merges*). Every s, i with $\text{vm_certified}(s) = \text{false}$ and $\text{vm_certified}(\text{vm_apply}(s, i)) = \text{true}$ has $\text{vm_apply}(\cdot, i)$ non-injective and $\text{instruction_cost}(i) \geq 1$ (*vm_certifying_step_is_priced_merge*). $\text{vm_apply}(\cdot, \text{JUMP } 0 \ 0)$ is not injective and costs 0 (*vm_jump_is_free_merge*). Hence $\neg \text{merging_steps_priced}(\text{vm_apply}, \text{instruction_cost})$ (*vm_prices_certifying_merge_leaves_others_free*). The schedule prices the certifying merge and leaves other merges free by design; A2 holds for the VM independently (*no_free_certification_certified*).

21.4 Exact Pricing Cannot Be Computed From a Window

Theorem 21.11 (No exact shadow price (*coq/kernel/nfi/ShadowPricing.v*), (S)). Let $\text{step} : S \rightarrow I \rightarrow S$, $\text{cert} : S \rightarrow \text{bool}$, and $\text{obs} : S \rightarrow O$ be arbitrary. A shadow price is $c(s, i) = \text{price}(\text{obs}(s), \text{obs}(\text{step}(s, i)))$. Call a collision two pairs (s_1, i_1) , (s_2, i_2) with equal observations before and after, where the first flips *cert* from false to true and the second does not. If a collision exists, every shadow price meeting the floor charges a non-flipping step (*shadow_floor_overcharges*), so no shadow price both meets the floor and never overcharges (*shadow_cannot_price_exactly*). The step price $\llbracket \text{flip} \rrbracket$ is exact (*step_price_is_exact*). If $\text{cert} = r \circ \text{obs}$ for some r , an exact shadow price exists (*window_showing_reading_prices_exactly*) and no collision exists (*window_showing_reading_has_no_collision*). On the VM, $(\text{abs_zero}, \text{CERTIFY } 0)$ and $(\text{abs_zero}, \text{JUMP } 1 \ 0)$ collide for *bare_observable*, and $(\text{abs_zero}, \text{CERTIFY } 0)$ and $(\text{abs_zero}, \text{JUMP } 1 \ 1)$ collide for *forget* (*vm_bare_observable*).

collision, vm_forget_collision), so neither window admits an exact price for VM certification (*vm_bare_shadow_cannot_price_exactly, vm_forget_shadow_cannot_price_exactly*). All nine theorems are closed under the global context.

21.5 Which Narrowing a Merge Price Charges

Theorem 21.12 (The machine’s spread is priced (*coq/kernel/nfi/KnowledgeNarrowing.v*), (S)). Let $\text{step} : S \rightarrow I \rightarrow S$ with decidable equality on S , and let cost meet *compression_priced*. For every duplicate-free list D and program t , with D' the duplicate-free list of states $\text{run}(s, t)$ for $s \in D$,

$$\lceil \log_2 |D| \rceil \leq \text{cost}(t) + \lceil \log_2 |D'| \rceil$$

(*run_narrowing_priced_log*). No decision tree or payment premise is supplied.

Theorem 21.13 (An observer’s knowledge is not priced (*KnowledgeNarrowing.v*), (S)). Let the observer’s knowledge after t from s_0 be the states of Ω whose observation along the run equals that of s_0 . On $S = \text{bool} \times \text{bool}$ with $\text{Measure}(x, d) = (x, d \oplus x)$ at cost 0 and $\text{Wipe}(x, d) = (x, \text{false})$ at cost 1, this particular cost meets *compression_priced*, and from $\Omega = \{(0, 0), (1, 0)\}$ the program $[\text{Measure}]$ at cost 0 leaves the display observer one candidate. Hence $\lceil \log_2 |\Omega| \rceil - \lceil \log_2 |\Omega'| \rceil \leq \text{cost}(t)$ fails for this admissible cost (*observer_narrowing_can_be_free*). Since *compression_priced* is only a lower bound, it may also admit costs that overcharge the injective measurement. The machine’s spread stays two (*demon_machine_spread_kept*), and every squeeze-priced cost charges *Wipe* at least 1 (*wipe_costs_at_least_one*). On the VM, *XOR_ADD 2 1 0* separates two states that differ in register 1 through register 2 at cost 0 (*vm_observer_narrowing_at_zero_cost*). All closed under the global context.

21.6 A Recursion Theorem for L

Theorem 21.14 (Kleene and Rice for L (*coq/kernel/foundation/LRecursion.v*), (S)). L is the weak call-by-value lambda calculus with de Bruijn terms and deterministic left-to-right reduction; $\lceil t \rceil$ is the Scott encoding. There is a closed Q with $Q \lceil t \rceil \rightarrow^* \lceil \lceil t \rceil \rceil$ (*Q_spec*). For every closed value s there is a closed t with $t \rightarrow^* s \lceil t \rceil$ (*second_recursion*). Every transformer computed on codes by a closed value has an extensional fixed point (*L_recursion_theorem*). No closed term decides, on codes, a property extensional in reached values that holds for one closed program and fails for another (*L_rice*); in particular halting (*L_halting_undecidable*). All closed under the global context. L is not a *Substrate* instance, since the interface requires a total *run*; the diagonal is restated for L .

21.7 The Dimensional Gap: Why Ledgers Cannot Fix Constants

Theorem 21.15 (Dimensional Gap (*coq/kernel/thermodynamic/DimensionalGapTheorem.v*), (S)). Let State be any type, $L : \text{State} \rightarrow \mathbb{N}$ any dimensionless integer ledger, $\omega : \text{State} \rightarrow \mathbb{N}$ any microstate-count function, and $S_{\text{cand}} : \text{State} \rightarrow \mathbb{R}$ any candidate entropy. Suppose for some base ≥ 2 and some state s with $L(s) \geq 1$:

1. **ledger-exponential:** $\omega(s') = \text{base}^{L(s')}$ for all s' ;
2. **ledger-linear:** $S_{\text{cand}}(s') = \alpha \cdot L(s')$ for all s' ;
3. **Boltzmann-form:** $S_{\text{cand}}(s') = k_B \ln \omega(s')$ whenever $\omega(s') > 0$.

Then α is forced:

$$\alpha = k_B \cdot \ln(\text{base}). \quad (21.1)$$

This is the sharpest honest statement of the limit on deriving physics from computation in this framework, and it deserves to be read as a **no-go**, not as a bridge. The theorem quantifies over *any* type, *any* integer ledger, and *any* exponential microstate count. It makes no reference to the Thiele machine at all. What it says is that a dimensionless integer ledger determines the entropy relation only *up to* two externally supplied scalars: the unit-conversion constant k_B (an empirical fact about thermodynamics) and the counting base (a structural fact about the substrate’s degrees of freedom). Fix both and α follows; fix neither and the ledger does not prefer any value. There is a one-parameter family of consistent (α, k_B) pairs, and choosing among them is precisely the modelling input that the ledger cannot supply.

Two consequences worth stating explicitly:

- The $k_B \ln 2$ appearing in the Thiele instantiation (`SecondLawBoltzmannWall.v`, base = 2 for binary cert-flips) is *not* derived from the machine. It is what you get after supplying k_B and the base from outside.
- Any claim of the form “this computational framework predicts a physical constant” is blocked by Theorem 21.15 unless it also says which two external scalars it assumed. This applies to the present development as much as to any other.

Read this section before the conditional physics chapters, not after: it is the constraint they operate under, and it is what keeps them honest.

Chapter 22

Self-Reference and the Thiele Manifold

Remark 22.1 (Scope: a definitional tower). Both files in this chapter stand alone. They import no kernel module and do not mention `VMState` or `vm_mu`. A `System` is a record of a natural number called `dimension` and a predicate on propositions. The meta-system is constructed with dimension one higher. The number 4, the name “spacetime”, and the `mu_cost` of a projection (a difference of two `dimension` fields) are chosen by definition. The chapter records a picture of self-reference escalating through levels and shows that the picture is consistent. It derives nothing about physical dimensions.

22.1 Self-Reference Escalation

Theorem 22.1 (Gödel-Style Escalation (`SelfReference.v`)). *Any self-referential system S requires a meta-system Meta with strictly more dimensions (in the record’s `dimension` field). The witness is `meta_system` S , defined with dimension $\dim(S) + 1$:*

$$\text{self_ref}(S) \implies \exists \text{Meta} : \dim(\text{Meta}) > \dim(S)$$

The meta-system inherits self-reference, creating an infinite escalation.

22.2 The Thiele Manifold

Definition 22.1 (Thiele Manifold (`ThieleManifold.v`)). An infinite tower of systems $\{L_n\}_{n \in \mathbb{N}}$ with:

- $\dim(L_n) = 4 + n$ (level n has dimension $4 + n$) in the canonical instance `canonical_manifold`. The record itself requires only $\dim(L_0) \geq 4$ and strict growth.
- $\dim(L_{n+1}) > \dim(L_n)$ (strict enrichment).
- Each level reasons about the level below.
- Self-reference at level n is answered by level $n + 1$.

Theorem 22.2 (Projection to Spacetime (`ThieleManifold.v`)). *The projection π_4 collapses the tower to dimension 4 (spacetime). For $n > 0$:*

- π_4 is lossy: $\dim(L_n) > 4$.
- $\mu\text{-cost of projection} = \dim(L_n) - 4 = n > 0$ in the canonical instance, and > 0 in general (`mu_cost_positive_for_projection`). This `mu_cost` is the file’s own dimension difference, not the VM ledger.

Spacetime is the “shadow” of the full manifold. That is the file’s reading of $\mathfrak{pi}4$ (named `spacetime_shadow`), not a theorem about physical spacetime.

Chapter 23

Three-Layer Observable Agreement

Remark 23.1 (Agreement, not isomorphism). No isomorphism is constructed in this chapter, and none is claimed: there is no pair of mutually inverse maps between the Coq, OCaml/Python, and RTL layers, and no proof that either composite is the identity. What is established is that each layer’s decoder satisfies a shared observable-projection contract, and that the three projections agree on every tested input: transitivity of equality through a common projection.

The distinction is not pedantic. Agreement on a projection is compatible with the layers differing arbitrarily on everything the projection forgets; an isomorphism would forbid that. Since the projection deliberately forgets internal representation, agreement is the right property to want here. Isomorphism would be a stronger claim than the construction supports.

Theorem 23.1 (Observable Comparison Contract (ThreeLayerIsomorphism.v)). *The active Coq theorem packages the observable-state contract used for cross-layer comparison: if the chosen Python and RTL decoders satisfy the same projection contract as the Coq layer, then the decoded observable states agree on that projection for a shared trace τ .*

$$\text{decode}_{\text{Python}}(\tau) = \text{decode}_{\text{Coq}}(\tau) \wedge \text{decode}_{\text{RTL}}(\tau) = \text{decode}_{\text{Coq}}(\tau) \implies \text{decode}_{\text{Python}}(\tau) = \text{decode}_{\text{RTL}}(\tau)$$

This is a contract theorem over explicit observable projections, not an unconditional claim that every current Python and RTL artifact is universally equal to Coq in all CI environments.

23.1 RTL Coverage: Zero Gaps

Theorem 23.2 (RTL gap registry and inventory (kami_hw/RTLGapRegistry.v), (S)). *rtl_gap_registry_empty: the hand-written RTL gap list is empty. rtl_inventory_arithmetic: $37 + 10 + 0 = 47$, the count of unconditional cases, cases under the joint representation invariant, and registered gaps among the 47 synthesized opcodes. Both are records, not coverage proofs: the first says what a list holds, and the second is arithmetic that mentions no opcode. The coverage evidence is the per-opcode theorems in the remark below and the step-level correspondence driven_step_wf under WFDDrivenPrecondition.*

Remark 23.2. The inventory is $37 + 10 + 0 = 47$. The 37 unconditional proofs are: 31 from the SupportedOpcode group, including CHSH_LASSERT (coq/kami_hw/EmbedStep.v: embed_step_compute) plus CALL, RET, CHSH_TRIAL (coq/kami_hw/EmbedStep_WF.v) and TENSOR_SET, TENSOR_GET, LASSERT (coq/kami_hw/GraphReconstructionBridge.v). The remaining 10 (PNEW, PSPLIT, PMERGE, MORPH, MORPH_ID, MORPH_DELETE, MORPH_ASSERT, MORPH_GET, COMPOSE, MORPH_TENSOR) carry Qed proofs under the joint structural invariant morph_table_wf $\wedge$ coupling_wf $\wedge$ coupling_desc_safe. Each component is preserved by every kami_step: morph_table_wf_kami_step_preserved, coupling_desc_safe_kami_step_preserved, and coupling_wf_kami_step_preserved. The

last takes `coupling_desc_safe` as a side hypothesis (in addition to `coupling_wf`), which is why the conjunction is the actual inductive invariant. So the precondition is discharged for any state reachable by machine execution. PNEW/PSPLIT/PMERGE additionally need `pt_well_formed` and arithmetic side conditions that are likewise inductive.

23.2 Named Trust Boundaries

The project has exactly two named trust boundaries. They are not Admitted proofs. They are explicit, auditable limits of the formal claim.

bsc_kami_compilation_trusted (Layer 2, Verilog semantics gap). The Bluespec SystemVerilog compiler generates Verilog from the Kami hardware specification. The correctness of the BSC compilation step, that the generated Verilog implements exactly the Kami spec, is trusted, not proved in Coq. This is documented in `coq/kernel/hardware_bridge/VerilogRTLCorrespondence.v` (Section 2b, “BSC Compilation Trust Boundary”) and the matching Definition `bsc_kami_compilation_trusted`. The Section Variable `rtl_step_correct` states that the synthesised RTL implements the Kami step function; this is the residual hypothesis that every theorem in the section quantifies over. The boundary is empirically backed by the cosim suite (`tests/test_verilog_cosim.py`, 38 tests, including `test_all_47_opcodes_in_cosim` which checks that every opcode appears) and the parametrized cross-layer fuzz suites (`tests/test_fuzz_random_programs.py` and `tests/test_cross_layer_adversarial_fuzz.py`).

A_QM (physical quantum axiom). Whether a physical Thiele Machine can produce CHSH-violating statistics by exploiting quantum entanglement in hardware is an open empirical question. The `column_contractive_iff_npa_psd` biconditional (Theorem 18.1) requires no physical axiom, it is pure algebra. The physical quantum axiom `A_QM` (declared as a `Context` parameter in `coq/PhysicsConditionalClosure.v`) is the separate named hypothesis that the chosen quantum CHSH experimental predicate admits a PSD completion with free within-party cross moments. It does not appear in the kernel-tier proof of the biconditional; it is the bridge used in conditional theorems such as `master_tsirelson_conditional`.

Chapter 24

Categorical Structure

24.1 The Morphism Layer

The active implementation adds a concrete category directly to the machine state. This is not an abstract analogy, it is an executable layer of 7 opcodes (0x27–0x2D) with formally verified laws.

24.1.1 Objects and Morphisms

The partition graph G carries two maps:

- `pg_modules` : $\text{ModuleID} \rightarrow \text{ModuleState}$, objects (partition modules)
- `pg_morphisms` : $\text{MorphismID} \rightarrow \text{MorphismState}$, arrows

A `MorphismState` bundles: source module, target module, coupling data (list of $(\text{nat} \times \text{nat})$ pairs plus a label), and an identity flag.

24.1.2 The Seven Categorical Opcodes

- `MORPH(dst, src, tgt, idx,  $\Delta\mu$ )`: create morphism $\text{src} \rightarrow \text{tgt}$, write morphism ID to `regs[dst]`. Costs $\Delta\mu$.
- `COMPOSE(dst, f, g,  $\Delta\mu$ )`: compose morphisms $f; g$ when $f.\text{target} = g.\text{source}$, write new ID to `regs[dst]`. Costs $\Delta\mu$. Errors if endpoints mismatch.
- `MORPH_ID(dst, m,  $\Delta\mu$ )`: create identity morphism id_m for module m (diagonal relation). Costs $\Delta\mu$.
- `MORPH_DELETE(f,  $\Delta\mu$ )`: remove morphism f . Cascades on module deletion via `graph_cascade_delete_morphisms`.
- `MORPH_ASSERT(f, prop, cert,  $\Delta\mu$ )`: cert-setter. Costs $S(\Delta\mu) \geq 1$ unconditionally, even for failed attempts (morphism does not exist). This is the No Free Insight law applied to relational claims.
- `MORPH_TENSOR(dst, f, g,  $\Delta\mu$ )`: tensor product $f \otimes g$. Requires that the union modules $\text{src}(f) \cup \text{src}(g)$ and $\text{tgt}(f) \cup \text{tgt}(g)$ already exist in the graph.
- `MORPH_GET(dst, f, sel,  $\Delta\mu$ )`: read field from morphism f : selector 0 = source, 1 = target, 2 = coupling length, 3 = `is_identity`.

24.1.3 Category Laws

Theorem 24.1 (Relational Composition is Associative (kernel/CategoryLaws.v), (S)). *For any three morphisms $f : A \rightarrow B$, $g : B \rightarrow C$, $h : C \rightarrow D$ with composable endpoints:*

$$(f;g);h = f;(g;h)$$

in the relational sense (coupling pairs of the composite). Zero Admitted.

Theorem 24.2 (Graph Composition Satisfies Laws (kernel/CategoryBridge.v), (S)). *The operation `graph_compose_morphisms` satisfies associativity and unitality. Additionally:*

$$\text{categorical_extension_nofi_consistent}$$

MORPH and COMPOSE can carry $\Delta\mu = 0$ and are not in the mandatory-cost class. The NoFI policy enters through MORPH_ASSERT: asserting that a built morphism chain exists and activating the cert channel always costs $S(\Delta\mu) \geq 1$. Building the chain is free; certifying it never is.

Theorem 24.3 (Tensor Bifunctionality (kernel/CategoryMonoidal.v), (S)). *`graph_tensor_morphisms` is a bifunctor: it satisfies the interchange law*

$$(f \otimes g);(h \otimes k) = (f;h) \otimes (g;k)$$

for composable, disjoint pairs.

24.1.4 The Categorical Separation Theorem

Theorem 24.4 (Categorical Separation (kernel/PartitionSeparation.v, §10), (S)). *There exist states s_1, s_2 such that:*

$$s_1.\text{vm_regs} = s_2.\text{vm_regs} \wedge s_1.\text{vm_mem} = s_2.\text{vm_mem} \wedge s_1.\text{vm_mu} = s_2.\text{vm_mu} \wedge s_1.\text{vm_err} = s_2.\text{vm_err}$$

but $s_1.\text{vm_graph.pg_morphisms} \neq s_2.\text{vm_graph.pg_morphisms}$.

Proof. Construct s_1 with `pg_morphisms` = $[(0, ms)]$ for any morphism state ms , and s_2 with `pg_morphisms` = $[]$. Arrange identical `vm_regs`, `vm_mem`, `vm_mu`, `vm_err` by choosing costs and instruction sequences accordingly. Distinction by `discriminate`. $\square$ $\square$

Remark 24.1 (Significance). A classical machine (exposing only registers, memory, and a cost counter) cannot distinguish s_1 from s_2 . The Thiele Machine distinguishes them in one probe instruction: `MORPH_DELETE 0` succeeds on s_1 and errors on s_2 . The categorical layer records *how* knowledge was derived, not just *what* the final values are.

This is the machine’s claim to being a *knowledge auditor*: any party claiming to have verified a structural relation must carry a μ -receipt $\geq$ the minimum construction cost. The NoFI theorem supplies a cost floor under its trace hypotheses. It does not authenticate an externally supplied receipt or prove arbitrary property text true.

Chapter 25

Falsifiable Predictions

The honest falsifiable predictions are the structural theorems that compile or fail. The claims below identify specific mathematical conditions that would constitute counterexamples to the machine’s core theorems.

25.1 Falsify No Free Insight

Find a Thiele Machine program trace that starts with `vm_certified = false` and $\mu = 0$, ends with `vm_certified = true`, and ends with $\mu = 0$. The Coq proof of `certification_requires_positive_mu` would fail.

25.2 Falsify the Classical CHSH Bound

Exhibit a locally factorizable strategy achieving $|S| > 2$. This contradicts `chsh_stat_violation_not_local` and also contradicts known mathematics.

25.3 Falsify the Biconditional

Exhibit $E_{00}, E_{01}, E_{10}, E_{11}$ satisfying the three column-contractivity conditions but with a non-PSD NPA moment matrix. The theorem `column_contractive_iff_npa_psd` says no such example exists.

25.4 Falsify the μ -Hierarchy

Find $k \geq 1$ and a trace reaching `vm_certified = true` and `vm_mu` $\geq k$ with total trace cost $< k$. This contradicts `mu_hierarchy_theorem`.

25.5 Open Questions

- **Physical interpretation of μ :** Whether the μ ledger corresponds to thermodynamic entropy in a physical Thiele Machine is an open empirical question. The formal development is consistent with Landauer’s principle. The calibration hypothesis is named (`mu_landauer_unruh_calibrated`) and not proved in Coq.
- **Quantum mechanics connection:** The mathematical equivalence between slice coherence and zero-marginal NPA realizability is a kernel-tier theorem. Whether a physical Thiele Machine can produce CHSH-violating statistics by exploiting quantum entanglement in hardware is an open empirical question outside the scope of this project.

- **Complexity implications:** Whether $\mu P \neq \mu NP$ is an open question. The factored-search witness shows a concrete structural advantage but is not a class separation.
- **Physical constants:** No physical constant is derived from first principles. There is no derivation of Planck’s constant h ; any relation tying an internal time-scale τ_μ to h would be circular by construction, defining τ_μ in terms of h and checking consistency rather than predicting h .
- **The two +1s (OP-Plus-One):** the +1 in `triangle_angle`’s denominator (`PI · d_bc / (1 + d_ab + d_ac + d_bc)`), `coq/kernel/curvature/MuGravity.v`) versus the A2 cost-floor +1 on `CERTIFY/LASSERT (coq/kernel/foundation/VMStep.v)`, same axiom, or unrelated regularisation? No kernel-internal theorem unifies them without circularity (`coq/kernel/frontier/F3_PlusOneStructural.v`); the symbolic correction term decays as $-\pi/(3(3d+1))$ in the equilateral case (`tools/triangle_angle_plus_one_analysis.py`), regularisation-shaped rather than A2-quantum-shaped. Inconclusive. A positive structural identification (Case A, the geometric +1 is A2 in geometric form) would re-frame the calibrated regime’s asymptotic-only character as a structural consequence of A2; a positive non-identification (Case B) would license replacing $S(\dots)$ with $(\dots)$ in the denominator. Either is real categorical content. The kernel was not modified pending closure.
- **F3 narrow non-vacuity** (related to OP-Plus-One): a constructible finite `VMState` satisfying $\forall m, \text{calibration_residual}(s, m) = 0$ on a partition graph with at least one actual graph triangle and `well_formed_triangulated` holding. The symbolic search to $N = 7$ exhausts negatively. Whether the asymptotic-only character of calibration is a structural consequence of OP-Plus-One’s Case A, or just a search-depth artifact, is itself open.

Appendix A

Proof File Index

The table maps theorem families to their Coq source files in the active build. Observation adequacy, joint event floors, and abstract calibration consistency are in `coq/kernel/frontier/ObservationPolicy.v`; descent from traces and its concrete targets are in `coq/kernel/frontier/TraceStateDescent.v`.

Result	Coq Source
<i>Cost ledger and No Free Insight</i>	
No Free Insight (Thm. 6.1)	<code>coq/nofi/NoFreeInsight_Theorem.v</code>
μ -Chaitin	<code>coq/nofi/MuChaitinTheory_Theorem.v</code>
μ -Conservation	<code>coq/kernel/foundation/MuLedgerConservation.v</code>
Receipt Integrity	<code>coq/kernel/nfi/ReceiptIntegrity.v</code>
Noether / Gauge	<code>coq/kernel/curvature/KernelNoether.v</code>
Strict Subsumption	<code>coq/kernel/foundation/Subsumption.v</code>
Universal NFI (any substrate)	<code>coq/kernel/nfi/UniversalCertificationCost.v</code> : <code>universal_nfi_any_substrate</code>
Quantitative NFI ($\geq S(d)$ trace bound)	<code>coq/kernel/mu_calculus/QuantitativeNoFI.v</code>
μ -Hierarchy	<code>coq/kernel/mu_calculus/MuHierarchyTheorem.v</code> : <code>mu_hierarchy_theorem</code>
Partition Ops Free, Certification Non-Free	<code>coq/kernel/nfi/PartitionRefinementNoFI.v</code>
μ -Ledger Necessity (5-way classification)	<code>coq/NecessityOfMuLedger.v</code> , <code>coq/kernel/nfi/NecessityAbstract.v</code>
Trace Cost for a Fixed Schedule	<code>coq/NecessityOfMuLedger.v</code> : <code>shadow_mu_is_computation_intrinsic</code>
μ Cost Universal Across Machines	<code>coq/NecessityOfMuLedger.v</code> : <code>shadow_mu_delta_universal</code>
μ Cost Unique Across Accounting Systems	<code>coq/NecessityOfMuLedger.v</code> : <code>shadow_mu_unique_accounting</code>
Positive-Cost Trace Accumulation	<code>coq/NecessityOfMuLedger.v</code> : <code>shadow_mu_inevitable</code>
μ -budget predicates + zero- μ inclusion	<code>coq/kernel/mu_calculus/MuComplexity.v</code>
<i>Categorical and shadow-projection layer</i>	
Categorical Separation / Category Laws	<code>coq/kernel/category/CategoryLaws.v</code> , <code>coq/kernel/category/CategoryBridge.v</code>
Shadow Strictly Lossy	<code>coq/kernel/witness/ShadowProjection.v</code> : <code>shadow_strictly_lossy</code>
Turing Strictness ($D5$)	<code>coq/kernel/foundation/TuringStrictness.v</code>
CHSH Coupling Bridge ($S \leq 2$ from local coupling, $S > 2$ rules it out)	<code>coq/kernel/quantum/CHSHCouplingBridge.v</code>
Witness Insight Three-Tier	<code>coq/kernel/witness/WitnessInsightGeneral.v</code>
Taxonomy	
<i>Quantum / CHSH algebra (no project-local axioms)</i>	

Slice-coherent $\iff$ NPA realizable (biconditional)	coq/kernel/quantum/QuantumPartitionPSD.v: column_contractive_iff_npa_psd
Forward: column-contractive $\Rightarrow$ NPA PSD	coq/kernel/nfi/MuLedgerQuantumBridge.v: zero_marginal_npa_column_contractive_implies_psd
Reverse: NPA PSD $\Rightarrow$ column-contractive	coq/kernel/quantum/QuantumPartitionPSD.v: npa_psd_implies_column_contractive
Quadratic-form discriminant lemma	coq/kernel/quantum/ConstructivePSD.v: quadratic_nonneg_discriminant
Tsirelson from column contractivity ($ S \leq 2\sqrt{2}$)	coq/kernel/nfi/MuLedgerQuantumBridge.v: state_column_contractive_implies_tsirelson (tsirelson_from_row_bounds in TsirelsonGeneral.v is the inner Cauchy–Schwarz step)
Tsirelson rational lower witness	coq/kernel/category/AlgebraicCoherence.v: tsirelson_rational_lower_witness
Tsirelson (algebraic from coherence)	coq/kernel/quantum/TsirelsonFromAlgebra.v
General Algebraic Tsirelson ($S^2 \leq 8$)	coq/kernel/category/AlgebraicCoherence.v: algebraically_coherent_tsirelson_general
Elliptope realizability (existential completion)	coq/kernel/quantum/ElliptopeCompletion.v: elliptope_realizable (definition); deterministic_strategy_elliptope, lhv_mixture_elliptope (every LHV correlator inside)
Tsirelson for the full elliptope ($S^2 \leq 8$)	coq/kernel/quantum/ElliptopeCompletion.v: elliptope_tsirelson (inner step psd_cauchy_schwarz)
PR box excluded; classical $\subset$ elliptope strict	coq/kernel/quantum/ElliptopeCompletion.v: pr_box_not_elliptope, beyond_classical_elliptope
Decidable elliptope gate (soundness)	coq/kernel/quantum/ElliptopeGate.v: elliptope_check_full_sound; boundary acceptance gate_accepts_pythagorean_boundary; universal refusal elliptope_full_gate_never_accepts_pr_box
<i>Frontier: pointer-observable criterion</i>	
Proliferation / unique-pointer schema	coq/kernel/frontier/PointerObservable.v: redundantly_proliferating, unique_pointer_among (definitions); toy_cert_unique_pointer (sanity instance)
Models of the five disciplines are unique pointers	coq/kernel/frontier/PointerObservableReductions.v: PoS_model_unique_pointer, Gas_model_unique_pointer, TEE_model_unique_pointer, CT_model_unique_pointer, PCC_model_unique_pointer; conjunction five_labeled_models_have_selected_pointer (closed under the global context). Correspondence to the deployed disciplines is argued in prose, not proved
Adversarial search: forgery-resistant systems that do <i>not</i> proliferate	coq/kernel/frontier/PointerObservableCounterexamples.v: deniable_authentication_model_not_proliferating, mac_model_not_proliferating, capability_model_not_proliferating; control public_log_model_proliferating. The Coq results fix the observer maps; (C1) and (C2) are argued in prose
<i>Conditional physics derivations (under named axioms; (C) tier)</i>	
Born Rule	coq/kernel/quantum/BornRule.v
Unitarity	coq/kernel/quantum/Unitarity.v
Purification	coq/kernel/quantum/Purification.v
No-Cloning	coq/kernel/quantum/NoCloning.v
Information Causality	coq/kernel/quantum/InformationCausality.v
Landauer derivation	coq/thermodynamic/LandauerDerived.v
Affine Off-Diagonal Ricci Zero (vertex 1)	coq/kernel/curvature/AffineEFEClosure.v
Full Tensor EFE at Vertex 1 ($G = 3T$)	coq/kernel/curvature/AffineEFEClosure.v

Outer Vertex Characterization (Ricci = 3/32)	coq/kernel/curvature/AffineEFEClosure.v
Spacetime Emergence (consistency check)	coq/kernel/curvature/SpacetimeEmergence.v
μ -Geometry (pseudometric)	coq/kernel/mu_calculus/MuGeometry.v
4D Simplicial Complex	coq/kernel/curvature/FourDSimplicialComplex.v
Metric from μ -Costs	coq/kernel/curvature/MetricFromMuCosts.v
Riemann / Christoffel / Einstein tensors	coq/kernel/curvature/RiemannTensor4D.v
Stress-Energy Tensor / Conservation / Vacuum EFE	coq/kernel/curvature/EinsteinEquations4D.v
Einstein Equations (general, full diagonal-and-offdiag chain)	coq/kernel/curvature/EinsteinEquationsFull.v
A_QM bridge axiom + master Tsirelson conditional	coq/PhysicsConditionalClosure.v
<i>Self-reference, falsifiability, isomorphism</i>	
Self-Reference Tower	coq/self_reference/SelfReference.v
Thiele Manifold	coq/thiele_manifold/ThieleManifold.v
Physical Constants (definitions only)	coq/thiele_manifold/PhysicalConstants.v
Falsifiable Prediction record	coq/kernel/aggregators/FalsifiablePrediction.v
Three-Layer (Coq $\leftrightarrow$ Python $\leftrightarrow$ RTL) Isomorphism	coq/kernel/hardware_bridge/ThreeLayerIsomorphism.v
<i>Hardware / RTL / extraction</i>	
RTL gap registry empty	coq/kami_hw/RTLGapRegistry.v: rtl_gap_registry_empty
RTL inventory count (37 + 10 + 0 = 47, arithmetic only)	coq/kami_hw/RTLGapRegistry.v: rtl_inventory_arithmetic
RTL Trust Boundary (Section Variable)	coq/kernel/hardware_bridge/VerilogRTLCorrespondence.v: rtl_step_correct
BSC Compilation Trust Boundary	coq/kernel/hardware_bridge/VerilogRTLCorrespondence.v: bsc_kami_compilation_trusted
Coq-Kami refinement (proves rtl_step_correct for COQKAMI)	coq/kami_hw/VerilogSemantics.v
EmbedStep / EmbedStep_WF (per-opcode bisimulation)	coq/kami_hw/EmbedStep.v, coq/kami_hw/EmbedStep_WF.v
GraphReconstructionBridge (TENSOR_*, LASSERT)	coq/kami_hw/GraphReconstructionBridge.v
ThieleCPU Kami specification	coq/kami_hw/ThieleCPUCore.v
OCaml extraction observable surface	coq/kernel/hardware_bridge/OCamlExtractionBridge.v: ocaml_observable_nofi_and_monotone

Appendix B

Epistemological Classification

The following table classifies every major result by its epistemological status.

Remark B.1 (How to read the tags). This table is the document’s principal credibility instrument, so four classifications that a reader is most likely to trip over are stated explicitly here.

- **Tsirelson** carries two entries that are not the same result. The *algebraic* bound is **(S)**: pure polynomial arithmetic, zero physics axioms. What is conditional is *physical realizability*, the **A_QM** bridge, which is its own line item and should never be folded into the algebra.
- **Born rule** likewise carries two entries for two results. *Born Rule Uniqueness* is **(C)**, conditional on the linearity assumption. The **BornRule.v** relation is **(R)**, because its μ -consistency hypothesis goes unused in the proof.
- **Affine EFE** is **(R)**: reference-only geometry. The containing chapter is discrete *Riemannian* geometry; the metric is positive-definite, so there is no causal structure to make a relativistic claim about (Remark 19.4).
- μ -**Chaitin** is **(C)**. It is a module-type contract with no instantiation anywhere in the corpus, and a contract with no instance is a conditional. Instantiating it is what would move it to **(S)**; the template is **Instance_Kernel.v**, which does exactly this for NFI.

Result	Status	Key Assumption
<i>Unconditional structural theorems about the machine</i>		
No Free Insight	(S)	Module type contract
μ -Chaitin	(C)	Module-type contract with <i>no instantiation anywhere in the corpus</i> ; the chain is four assumed inequalities. Conditional until instantiated (cf. Instance_Kernel.v for the NFI template)
Strict Subsumption	(S)	Partition structure
Oracle Cost (instantiation of universal NFI)	(S)	Definitional (soundness $:= \text{cost} \geq 1$)
μ -Conservation	(S)	Machine semantics only
μ -Ledger Necessity	(S)	Machine semantics + explicit witnesses
μ -Ledger Irredundancy (neither μ nor cert droppable)	(S)	Quantifies over all carriers/projections. <i>Not</i> minimality: no order on projections is defined, so nothing is proved least (theorem P_full_complete_neither_mu_nor_cert_droppable)
Receipt Integrity	(S)	Hash-chain model
Gauge / Noether	(S)	Definitional (record structure)

Three-Layer Observable Agreement (<i>not</i> an isomorphism: no pair of mutually inverse maps is constructed; the content is transitivity of equality on a shared projection)	(C)	Backend decoders satisfy the shared observable projection contract
Discrete Angle-Defect Identity (planar triangulations)	(S)	Well-formed triangulated partition graph; equilateral model; not classical Gauss-Bonnet
Curvature from μ -Gradients (uniform mass $\Rightarrow G_{\mu\nu} = 0$)	(R)	Reference only; $0 = 0$ in vacuum
Vacuum Implies Zero Stress-Energy	(R)	Reference only; $0 = 0$ in vacuum
Einstein Equations (Vacuum)	(R)	Reference only; not a claim about physics; $G = 1/(8\pi)$ is a definition
Einstein Equations (General full tensor)	(C)	Named off-diagonal Ricci premise in EinsteinEquationsFull.v
Affine Off-Diagonal Ricci Zero (vertex 1)	(R)	Reference only; specific boundary 4-simplex configuration
Full Tensor EFE at Vertex 1 ($G = 3T$)	(R)	Reference only; not a physical claim
Outer Vertex Characterization	(R)	Reference only
Discrete Differential Geometry	(R)	Definitions only; reference exploration
Partition Ops Free / Certification Non-Free	(S)	Machine semantics only
CHSH Trial Is Tier 0	(S)	Machine semantics only
Three-Tier Observation Taxonomy	(S)	Machine semantics only
Locally Consistent $\Rightarrow$ Separable Coupling	(S)	Coupling definition
Local Coupling Bound ($ S \leq 2$)	(S)	Coupling definition
$S > 2$ Rules Out Local Couplings	(S)	Coupling definition
RTL Bisimulation (driven_step_wf)	(S)	Per-opcode Kami/ vm_apply agreement under WFDrivenPrecondition . (rtl_inventory_arithmetic , $37+10+0 = 47$, is Peano arithmetic and is <i>not</i> coverage evidence)
Slice-Coherent $\iff$ NPA Realizable	(S)	Polynomial arithmetic; standard-library assumptions
PR Box Not Slice-Coherent	(S)	Direct arithmetic on condition 1
Elliptope Realizability (full correlator set)	(S)	Existential completion; stdlib reals only
Decidable Elliptope Gate Sound	(S)	Fraction-free $\mathbb{Z}$ arithmetic, two branches
Five Disciplines' Models Are Unique Pointers	(S)	Closed under the global context; model structure only. Evidence for (PO-PUBLIC) only through the prose correspondence; see Section 18.3.1
Strong Pointer Criterion (PO-STRONG) Is False	(S)	Coq checks the three observer maps (deniable authentication, symmetric MAC, object capabilities); their faithfulness to the real designs, and so the refutation, is argued in prose
Tsirelson Is Thiele Boundary (forward + contrapositive)	(S)	From column contractivity; converse not claimed
Zero- μ traces satisfy the preservation budget	(S)	Error-free traces; no returned-answer or uniform-solver requirement
Forced price is merging; merge or revoke	(S)	Finite enumeration; decidable instruction equality for the iff; converse to permanence fails on three states

Entropy drop of a permanent flip	(S)	Finite enumeration, permanent certificate; Shannon entropy over Coq reals; standard real-number axioms only
Finite VM instance	(S)	Eight-state machine; VM fragment <code>CERTIFY 0, JUMP a 2, CHECKPOINT</code> through the window <code>(pc mod 4, certified)</code>
Full VM prices only the certifying merge	(S)	Every certification flip is a priced merge; <code>JUMP 0 0</code> is a free merge; by design
No exact shadow price	(S)	Any window with a collision; VM collisions for <code>bare_observable</code> and <code>forget</code>
<i>Conditional derivations (valid given stated axioms)</i>		
Born Rule Uniqueness	(C)	Linearity assumption (<code>coq/kernel/quantum/BornRule.v</code> , <code>BornRuleLinearity.v</code>)
Unitarity from $\mu = 0$	(C)	Info conservation and purity-nonincreasing premises; converse not proved
No-Cloning	(C)	Purity = information; conservation inequality supplied as a premise
Purification	(C)	Bloch sphere model
Landauer	(C)	Second law <i>assumed</i> as record field
A2 from a merging price	(C)	Finite enumeration, permanent certificate, non-injective steps cost ≥ 1 (Landauer premise); core merge theorem is (S)
Cost of a squeeze	(C)	Counting premise: cost c allows at most a 2^c -fold squeeze; bound attained by a four-state stamp
Heat floor of a permanent flip	(C)	Landauer's principle as the named premise <code>landauer_heat</code>
Self-Reference	(C)	Dimension = complexity; the dimension is a record field chosen by definition
<i>Consistency relations (definitions verified, not predictions)</i>		
Information Causality	(R)	Circular: equivalence assumed as hypothesis
μ -Geometry	(R)	Pseudometric (absolute value on $\mathbb{Z}$)
Spacetime Locality	(R)	Data isolation in partition graph
Born Rule (<code>BornRule.v</code>)	(R)	μ -consistent hypothesis unused

Legend: (S) = unconditional theorem about the machine; (C) = valid derivation conditional on stated axiom; (R) = internal consistency check.